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RESEARCH ARTICLE

Dynamic Real-Time Anchor Selection for Accurate UWB Indoor Positioning-Based Deep Neural Networks

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ABSTRACT In wireless localization systems, enhancing location estimation performance is critical, particularly in challenging environments, such as military urban operations and emergency response scenarios. Ultra-wideband (UWB) positioning systems using two-way-ranging (TWR) schemes avoid synchronization issues, but face challenges related to scalability, anchor selection, and poor channel characteristics. Existing methods often rely on exhaustive geometric calculations, leading to inefficiencies in dynamic and time-critical scenarios. This paper proposes a lightweight sequential branching deep network for dynamic indoor positioning (LSB-DIP Net) is proposed to address these challenges. By integrating multi-scale feature extraction, sequential learning, and advanced activation functions with the conventional linearized least squares method, the LSB-DIP Net enables robust, accurate, and dynamic UWB positioning in real-time. The model effectively mitigates non-line-of-sight (NLOS) ranging errors, evaluates anchor channel quality online, and selects optimal anchor combinations, ensuring scalability and adaptability for diverse deployment scenarios. The proposed approach demonstrates exceptional performance in dynamic setups, achieving low mean squared error (MSE) of 0.0051m^2, high accuracy in identifying anchor channels of 99.44%, with a maximum positional error of less than 0.17 m in harsh environments. Validated across public datasets, the system ensures generalizability and outperforms state-of-the-art counterparts in the market, making it a reliable tool for real-time applications in communication and navigation systems.

INDEX TERMS Deep neural network (DNN), dynamic indoor positioning systems (IPs), anchor selection (AS), long short-term memory (LSTM), robotics.

I. INTRODUCTION

Indoor positioning systems (IPSs) aim to provide efficient, accurate, and adaptable localization solutions for dynamic and expansive indoor environments. Unlike outdoor positioning technologies such as GPS, which excel in open spaces, indoor environments present significant challenges due to structural obstructions, multipath signal propagation, and attenuation. These challenges have underscored the need for specialized indoor localization techniques, particularly as the demand for high-precision applications in environments such as commercial buildings, airports, and museums continues to grow. IPSs technologies have been applied across a wide range of domains, enhancing operational efficiency, navigation, and user experiences [1], [2].

To address the requirements of indoor localization, such as accuracy, latency, cost, and adaptability, various technologies, such as smartphone sensors [3], Wi-Fi received signal strength indication (RSSI) signals [4], cameras [5], Bluetooth [6], and ZigBee [7], have been developed. However,

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the localization accuracy of certain technologies is often limited and can exceed 2.5 meters, with significant performance degradation in NLOS conditions or in the presence of multipath propagation due to signal reflections. For instance, Wi-Fi RSSI-based positioning is highly susceptible to signal interference and multipath fading. Meanwhile, Bluetooth and ZigBee are cost-effective options; however, they have limited penetration capabilities and are highly affected by environmental changes. These limitations make these technologies unsuitable for applications that require precise localization in complicated indoor environments.

In contrast, advanced methods such as UWB and vision-based approaches can achieve sub-meter accuracy under controlled conditions. Such limitations make IPSs insufficient for high-precision applications in complex indoor environments. Interestingly, UWB technology with its unique characteristics, including wide bandwidth, low power use, high penetration capabilities, and robustness to multipath interference, has emerged as a promising solution to address challenges facing IPSs [8], [9]. The fine time resolution of UWB enables the use of time-of-flight (TOF) techniques, resulting in centimeter-level localization accuracy. Unlike RSSI-based methods, which suffer from interference and fading, UWB provides reliable distance estimation by leveraging precise signal timing and immunity to interference from other wireless systems [10], [11]. Therefore, UWB has become a preferred choice for high-precision localization applications in dynamic and complex indoor environments [9].

Indoor environments might be divided as structured or known, semi-structured, and unstructured or unknown based on the control that the IPSs takes possession of them [12], [13]. In wireless communication systems, indoor positioning algorithms are broadly classified into range-based and range-free approaches. In a range-based method, the location of a node is determined by comparing it to the positions of other nearby nodes. Before attempting to determine the position of the unknown node, it is necessary to accurately estimate the angle or distance (range information) between the anchors. RSSI [14], TOF [15], time of arrival (TOA) [16], time difference of arrival (TDOA) [17], and angle of arrival (AOA) [18], and other technologies are used to achieve high accuracy. However, these technologies often involve significant computational complexity and implementation costs. Range-free approaches, on the other hand, rely on network topology and connectivity information rather than direct signal metrics, offering simpler implementation at the cost of reduced accuracy [20]. In UWB systems, range-based techniques, such as trilateration, multilateration, and AOA, are widely used for their ability to derive precise location estimates by analyzing signal propagation times [19].

Despite these advantages, UWB systems face notable challenges, particularly in NLOS conditions where obstacles obstruct the direct path between the anchor and the mobile device. NLOS propagation introduces biases in distance estimation, leading to degrading localization accuracy. Existing mitigation strategies often involve computationally intensive preprocessing or require dense anchor deployments to improve performance, which increases system complexity, energy loss, and deployment costs [21], [22], [23], [24]. However, these methods demand extensive laboratory work and involve complex preprocessing techniques, which can significantly increase computational complexity and latency, further limiting their generalizability, particularly in challenging and dynamic indoor environments. Additionally, deploying many UWB anchors to improve localization accuracy can be energy-intensive, limiting the scalability of such systems. Therefore, these limitations necessitate the development of efficient anchor selection strategies to enhance performance while maintaining scalability and resource efficiency.

To address these challenges, this study introduces a lightweight sequential branching deep network for a dynamic indoor positioning-based anchor selection (AS) strategy. The key contributions of this work are summarized as follows:

1) Development of an adaptive anchor selection strategy (LSB-DIB Net), which dynamically ranks and selects anchors based on real-time channel quality classification, optimizing positioning accuracy and reliability.
2) Implementation of a multi-class NLOS classification framework, enhancing error mitigation and reducing positioning biases compared to conventional binary classification approaches, including line-of-sight (LOS), soft non-line-of-sight (soft-NLOS), hard non-line-of-sight (hard-NLOS), and multipath (MP).
3) Comprehensive experimental validation, utilizing real-world datasets and diverse test environments to demonstrate the robustness and generalizability of the proposed model.
4) Optimization of computational efficiency, ensuring a low-complexity design while maintaining real-time processing capabilities for practical deployment in UWB-based indoor positioning systems.

Addressing critical challenges such as NLOS mitigation, scalability, and adaptability, this study advances the field of UWB-based indoor positioning and offers a robust solution for real-world applications.

II. RELATED WORK

Since the emergence of UWB wireless communication technology, recent research efforts have focused on developing accurate indoor positioning systems. In particular, UWB-based TWR positioning systems have been a subject of significant interest, primarily focusing on techniques for mitigating ranging errors to achieve high-precision localization in indoor environments. Where a positioning algorithm estimates the two-dimensional or three-dimensional coordinates of a mobile node by utilizing distance measurements to a set of fixed reference points, known as anchors. Anchor node selection in indoor localization aims to enhance network

performance by strategically selecting a subset of beacons based on specific criteria. A critical consideration in indoor localization is the mitigation of location errors caused by NLOS propagation, especially in real-time with dynamic environments. This section examines state-of-the-art algorithms related to anchor selection. Presently, there are limited approaches to anchor selection algorithms, to our knowledge, available for TWR-UWB localization systems. Furthermore, although anchor selection algorithms form the foundation of our proposed methodology, recent advancements in alternative positioning techniques over the past two years have also been reviewed and discussed at the end of this section.

UWB-channel impulse response (CIR) was proposed by Sangmo et al. 2023 to obtain enhanced NLOS probabilities based on deep learning (DL) [25]. The proposed UWB-CIR method integrates DL with Kalman filtering to mitigate ranging errors in NLOS conditions, thereby enhancing accuracy in distance estimation and trilateration-based positioning. Experimental results from real-time settings, including open spaces, pockets, and bags, were implemented to validate the effectiveness of the proposed weighted adaptive Kalman filter (WAKF) method. The WAKF achieved an accuracy within a ± 50 cm tolerance compared to conventional methods. However, challenges persisted in highly complex environments with dynamic and varied obstructions, as the DL model currently classifies only a limited range of NLOS scenarios (pocket, bag, and hand). In particularly complex indoor settings, such as those with multiple structural partitions or highly cluttered industrial environments, this method may encounter unforeseen positioning errors.

Extended Kalman Filter (EKF) as a non-linear state estimator was proposed by Poulose et al., 2020 to reduce localization errors with an increasing number of anchors [26]. This algorithm refines user position estimates based on TOA measurements from each anchor, effectively accounting for system model non-linearities. Simulations were conducted to evaluate the estimator's efficacy in a two-dimensional environment, examining various anchor configurations and assessing performance under both LOS and NLOS conditions. Results indicated that, as anchor numbers increase, the EKF-based approach sustains positioning accuracy while reducing computational complexity more effectively than conventional methods used. The UWB-EKF estimator was also benchmarked against conventional localization algorithms, including linearized least squares estimation (LLSE), weighted centroid estimation (WCE), and maximum likelihood estimation (MLE), demonstrating an obtained accuracy within a ± 50 cm tolerance range. While the EKF presents a robust solution for mitigating NLOS effects in controlled simulations, its efficacy may be constrained in highly obstructed or complex environments. Additionally, the EKF's dependence on specific noise characteristics could limit adaptability to varying interference or unpredictable noise profiles, potentially diminishing accuracy in real-world applications.

A methodology that uses an anchor selection approach to enhance UWB-based indoor positioning accuracy without incorporating NLOS mitigation techniques was proposed by Albaidhani and Alsudani, 2021 [27]. Anchor nodes were randomly deployed within the indoor environment and clustered into groups. Each group independently estimates the mobile station's position using the linearized least squares (LLS) method. The accuracy of each group's estimate is evaluated using the Weighted Geometric Dilution of Precision (WGDOP) metric, which considers both distance variance and spatial distribution. The group exhibiting the lowest WGDOP value, indicative of the highest accuracy, is selected. Subsequently, a modified LLS approach is applied to further refine the position estimate. This methodology streamlines system complexity while improving accuracy, achieving an MSE of approximately 25 cm^2 in experimental validations across two trajectories. However, the proposed system, while promising, exhibits limitations in complex indoor environments. The omission of NLOS identification and mitigation, though simplifying the system, can compromise accuracy in scenarios with frequent signal obstructions. Additionally, the reliance on random anchor placement and WGDOP-based clustering may not guarantee optimal spatial distribution in dynamic or irregular spaces, potentially affecting positioning precision. The system's performance is also contingent on the specific environmental conditions tested, raising concerns about its generalizability and robustness in diverse real-world settings. Furthermore, the controlled experimental validation may not comprehensively reflect the complexities inherent in real-world indoor environments, which could limit its practical applicability in scenarios requiring high accuracy within challenging indoor settings.

An approach that aims to enhance the accuracy and efficiency of UWB indoor positioning systems through a novel AS algorithm was proposed by Albaidhani et al., 2019 [28]. It begins by deriving the MSE to estimate the mobile station's (MS) position via the linearized Least Squares (LS) method. This MSE calculation serves as the core metric for assessing the accuracy of various anchor node configurations, enabling the system to dynamically identify the configuration that offers the highest precision in real-time. The AS algorithm evaluates potential groups of anchor nodes and selects the combination that minimizes positioning error, as measured by MSE. To increase the system's adaptability, a "virtual node" concept is integrated, allowing the AS algorithm to maintain high accuracy even when only two anchor nodes are available by incorporating this virtual node into the linearized LS calculation. Once the optimal anchor group is selected, the MS position is recalculated using the Weighted Least Squares (WLS) method, which further refines the position estimate by accounting for variations in measurement accuracy among the selected anchor nodes. The proposed method was tested in controlled indoor

environments, where it achieved a positioning accuracy with an MSE of 50 cm^2. However, the proposed method, while demonstrating potential, faces several limitations. Firstly, the computational overhead associated with the real-time evaluation of multiple anchor configurations and the iterative calculation of MSE and WLS may restrict its applicability in low-latency or resource-constrained applications. Secondly, the method's evaluation was primarily conducted in controlled indoor environments, raising concerns regarding its performance in complex real-world scenarios characterized by extreme clutter or dynamic conditions. Lastly, the reliance on a locally collected dataset limits the generalizability of the method's accuracy across different devices, datasets, and application contexts, necessitating further validation in diverse settings.

Another methodology that performs the location predictions without relying on NLOS identification methods based on the jump-start reinforcement learning (JSRL) approach was proposed by Zohreh et al., 2022 [29]. A combination of the guide and exploration policies was used to speed up the early learning phase and improve the ability of the pre-trained RL frameworks. The JUNO framework, grounded in JSRL, introduces an anchor node selection method that empowers the mobile user to autonomously identify optimal LOS connections within UWB indoor localization systems. The state space is discretized for computational efficiency, and actions involve selecting optimal UWB beacons, with a reward function focused on minimizing localization errors. The model employs a guide-step and exploration-exploitation balance, controlled by an epsilon parameter, to optimize learning and decision-making. This approach uses location error and cumulative rewards as evaluation metrics to assess the performance of the proposed JUNO framework. The JUNO framework achieves positioning accuracy with a localization error of 32 cm^2. However, the simulation results were conducted under controlled conditions, assuming a uniform distribution of obstacles and NLOS conditions. This idealization may not accurately capture the variability and irregularity of real-world indoor environments. Consequently, the model's robustness and generalization capabilities in larger or more complex settings might be limited, necessitating extensive evaluation across diverse scenarios.

The Best Anchor Selection for Trilateration (BAST) algorithm was also proposed by Courtay et al., 2019 [30]. This method initially filters erroneous TOF measurements and employs a linear forward predictor to estimate the mobile node's position based on a constant speed model. The algorithm iteratively selects optimal anchor triplets by calculating the Euclidean distance (ED) between predicted and calculated positions, choosing the triplet with the minimal ED. Noise estimation is performed by comparing measured and actual distances from the selected triplet, with the noise vector subsequently subtracted from the measured distances. Final trilateration then determines the precise coordinates of the mobile node. While BAST achieved promising results in serpentine and tennis court scenarios, with average errors of 0.06 and 1.08 meters, respectively, its evaluation is limited to these specific settings. This raises concerns regarding its generalizability to diverse indoor environments, particularly those with complex layouts, multiple levels, and significant obstacles. Additionally, the method's sensitivity to TOF measurement noise and its potential limitations in handling NLOS propagation effects, especially during rapid movements, may impact its performance. Furthermore, the computational complexity of BAST may hinder its suitability for real-time applications or resource-constrained devices. Finally, the lack of discussion on the algorithm's generalizability to different settings and datasets limits our understanding of its robustness and applicability in diverse conditions.

Two anchor selection algorithms for TWR-UWB-based indoor positioning systems were recently introduced by Herbruggen et al., 2024 [31], aimed at enhancing localization accuracy and update rates. The first approach is a non-constrained algorithm that communicates with all available anchors to collect comprehensive link quality metrics. It subsequently selects a subset of anchors based on both link quality and geometric dilution of precision (DOP) to improve positioning accuracy. The second approach is a real-time, lightweight algorithm implemented on the UWB tag, designed to operate under limited computational resources. This method continuously monitors link quality and dynamically replaces underperforming anchors to ensure efficient and accurate localization, without relying on external position or anchor coordinate information. Both algorithms were validated in a realistic industrial environment. Furthermore, the localization server utilizes a Kalman filter to combine TWR-measured distances and anchor coordinates, producing a refined position estimate. The non-constrained algorithm demonstrated an accuracy of 14.9 cm, while the on-tag real-time algorithm achieved an accuracy of 20.2 cm. However, despite its high accuracy, the non-constrained algorithm demands substantial computational resources and real-time data from all available anchors, making it unsuitable for low-power, resource-constrained devices. This limitation poses challenges for large-scale or energy-sensitive deployments. Additionally, both algorithms depend on specific link quality metrics, which may not generalize effectively to diverse or more complex indoor environments, particularly those with severe NLOS conditions or fluctuating interference patterns. Moreover, the proposed methods have only been evaluated in a controlled industrial setting, raising concerns about potential performance degradation in less predictable or more dynamic real-world scenarios.

In another study, a framework that combines channel-quality assessment with adaptive anchor node selection to enhance positioning accuracy [32]. Initially, the evaluation process incorporates a series of signal preprocessing techniques, including noise filtering and peak detection, implemented using the ordered statistical constant false alarm rate (OS-CFAR) method. OS-CFAR dynamically establishes a

detection threshold to distinguish significant signals from background noise, ensuring that only reliable CIR data is utilized. From the preprocessed CIR, three key metrics are extracted to assess channel quality. The ideal channel factor (A) quantifies the resemblance between the current CIR and a reference CIR obtained under ideal LOS conditions. The power dispersion factor (B) represents the ratio of power in the first path to the total received power, where higher values denote better channel conditions. The first path intensity factor (C) evaluates the delay between the first path and the strongest path in the CIR, with shorter delays indicating fewer NLOS effects. These three metrics are subsequently combined using a weighted equation to derive a comprehensive channel quality score (Q), representing the overall reliability of the measured distance. Following this, the anchor node selection process leverages channel quality scores to identify the most reliable anchor nodes for positioning. The framework assumes a setup with surplus anchor nodes, allowing the selection of nodes with superior quality scores while discarding data from nodes affected by NLOS or multipath interference. If the overall channel quality is deemed high, anchor nodes are additionally selected based on the DOP. Finally, the chosen anchor nodes are employed within a least-squares estimator to determine the target's position. The proposed approach is validated across various environments, including offices, corridors, and halls, with results indicating a reduction in RMSE achieving up to a 95.78% improvement in specific cases and an average reduction of 40.4%. While the proposed method effectively addresses certain NLOS scenarios, its performance may degrade in extreme multipath environments, where accurately distinguishing between LOS and NLOS conditions becomes challenging. Additionally, the channel quality evaluation relies on empirically determined weights for the ideal channel factor, power dispersion factor, and first-path intensity factor. The effectiveness of these weights may vary across different environments, potentially limiting the method's adaptability and necessitating recalibration. Furthermore, the multi-step nature of the channel quality evaluation process, involving noise filtering, OS-CFAR detection, and CIR-based metric calculation, introduces computational overhead, which may be impractical for real-time applications or resource-constrained devices.

The authors of [33] presented an indoor positioning system based on Long Short-Term Memory (LSTM) neural networks and a two-stage activity classification approach. The system employs inertial sensors integrated within smartphones to identify physical activities (such as walking, running, stationary periods) and action units (AUs), including left/right turns and variations in step length. The system architecture comprises two LSTM models dedicated to activity and AU classification, respectively, coupled with a moving distance estimator for trajectory prediction. This approach addresses challenges associated with step detection errors and indoor localization inaccuracies, achieving an average localization error of 0.782 meters, without requiring complex hardware deployments or reliance on RSSI-based methodologies. However, the system's reliance on mode values derived from prior distributions may not fully capture real-time fluctuations in step length resulting from unforeseen alterations in movement patterns. Furthermore, further evaluation in diverse and challenging indoor environments is necessary to mitigate concerns regarding generalizability in complex indoor settings.

The study presented in [34] focuses on enhancing indoor positioning accuracy using UWB technology by incorporating LSTM neural networks. The proposed system employs LSTM networks to predict ranging errors based on CIR signals, enabling more accurate error correction without explicit differentiation between LOS and NLOS conditions. WLS and regression-weighted least squares (WRLS) algorithms are utilized to optimize positioning performance. With a configuration of eight anchors, the system achieved positioning errors of 0.74 m, 0.86 m, and greater than 1 m for WLS, WRLS, and LS, respectively. However, the experimental evaluation is limited to a specific dataset, which may restrict the generalizability of the findings to other environments. Furthermore, the positioning accuracy is susceptible to variations in both the distribution and number of deployed anchors.

The work presented in [35] investigates an indoor localization system employing UWB technology in conjunction with LSTM neural networks. The system enhances the accuracy of user position estimation by utilizing TOA data processed through LSTM networks. This hybrid methodology integrates the benefits of UWB's high-precision signal characteristics with LSTM's capability to model temporal dependencies. The research incorporates simulations of a trilateration algorithm and evaluates the proposed system's performance under varying hyperparameter configurations. The method addresses LOS challenges, achieving a mean error of approximately 7 cm. However, the reported performance is derived from evaluations conducted within a controlled LOS environment, lacking the complexities of real-world indoor settings characterized by NLOS conditions. Consequently, the generalizability of these results to real-world scenarios with significant obstructions or interference may be limited. Furthermore, the achieved accuracy is contingent upon the number and placement of UWB anchors, potentially restricting scalability to larger spaces. Additionally, the relatively high training time required for the LSTM model could pose challenges for real-time deployment.

An integrated indoor positioning system employing a loosely coupled fusion architecture of inertial navigation systems (INS) and UWB technology was proposed by Cheng and Fu, 2024. An adaptive extended Kalman filter (AEKF) is implemented, which adaptively adjusts processes and measures noise covariance matrices, replacing the conventional EKF. Motion tracking is refined through a DL-based regression model, utilizing a recurrent neural network (RNN) to estimate step length and heading direction from tri-axial accelerometer and gyroscope data. Furthermore, particle

filtering is augmented via a firefly algorithm-enhanced particle filter (FAPF) to mitigate particle impoverishment and improve convergence efficiency. A NLOS compensation mechanism, leveraging IMU-predicted motion data for UWB distance estimation correction, is integrated. The framework also incorporates an adaptive threshold gait detection algorithm, eliminating the necessity for pre-defined thresholds in zero-velocity updates (ZUPT), thereby improving pedestrian motion tracking precision. The system achieves an average localization error of 40 cm, with RMSE of 6.49 cm in LOS and 31.64 cm in NLOS environments, utilizing a publicly available UWB dataset. However, the computational complexity is increased due to the firefly algorithm optimization, rendering it more resource-intensive compared to traditional particle filtering methods. The deep learning regression model for step length and heading estimation necessitates extensive training data, and inaccuracies in these predictions can adversely affect overall positioning accuracy. Moreover, while the NLOS compensation mechanism enhances UWB localization accuracy, its efficacy is reduced in densely cluttered environments characterized by significant signal obstruction [36].

A Dirichlet process gaussian mixture model (DPGMM) to adaptively model and estimate channel bias was proposed by Li et al., 2025. By treating channel bias as a stochastic variable, the proposed method leverages Bayesian inference to learn its probabilistic distribution. Position estimation is refined through the implementation of a maximum A posteriori (MAP) estimator, which integrates ToF and AoA measurements. Furthermore, a leave-one-out cross-validation approach is utilized to iteratively improve measurement accuracy by incorporating observations from multiple UWB anchors. The system is implemented using single-anchor, omnidirectional transmitting antennas, with a motion capture system providing ground-truth trajectory data. In an open indoor environment, the system achieves RMSE of 29 cm in LOS conditions and 37 cm in NLOS conditions (albeit constrained in NLOS scenarios with metal plates). However, the bayesian learning framework, while enhancing accuracy, introduces computational overhead, resulting in increased demands relative to conventional UWB localization approaches. Additionally, while single-anchor localization is supported, optimal performance is contingent upon a multi-anchor configuration, where multiple observations contribute to improved accuracy [37].

An Enhanced LLS (ELLS) method is designed by Kateb et al. 2023 to improve positioning accuracy while simultaneously reducing computational complexity and minimizing the number of required anchor nodes [38]. The proposed methodology is predicated on a modified variant of the LLS estimation technique, incorporating a variance-based refinement process to enhance the precision of position estimation. The system initiates by deploying four randomly positioned UWB anchor nodes and measuring the distance between the mobile station and each anchor. Furthermore, it integrates GDOP optimization, which selects the optimal anchor node configuration based on signal strength and measurement consistency. The system's performance was evaluated under two distinct scenarios, each encompassing varying propagation channels, including LOS, soft-NLOS, and hard NLOS conditions. The results demonstrate that the proposed ELLS method achieves a MSE of 0.46 m^2. However, the method's scalability to large-scale environments remains unexplored, and its performance may be compromised in highly dynamic indoor settings characterized by frequent object and obstacle alterations. Additionally, while random anchor node placement is supported, the efficacy of the approach may still be influenced by the spatial distribution of the anchors, indicating that suboptimal placement could result in diminished accuracy.

Another approach for improving indoor positioning accuracy is introduced by Huang et al. [39], which integrates particle swarm optimization (PSO) with a multi-anchor UWB system. The proposed system adapts dynamically to varying indoor conditions by optimizing anchor selection and adjusting the impact of NLOS-affected anchors during the position estimation process. The methodology involves deploying a three-dimensional multi-anchor UWB positioning system, where six UWB anchors are placed in predefined locations. Four anchors operate in LOS conditions, while two are positioned in NLOS environments to evaluate the impact of signal obstruction. The initial position of the target device is estimated using an LS method, solving a matrix-based equation derived from distance measurements between the target and anchors. The study employs PSO, an iterative optimization algorithm, which dynamically adjusts NLOS anchor weights, ensuring that distance measurements from obstructed anchors contribute less to the final position computation. The PSO algorithm operates in a 3D search space, where each particle represents a potential position estimate. The algorithm iteratively updates position estimates based on previous best positions, converging toward the optimal solution over 75 iterations. Experimental results demonstrate that they reduced the error in NLOS scenarios from 33 cm to 10 cm. However, these results are contingent upon the idealized scenario of four anchors operating exclusively under LOS conditions, a configuration rarely achievable in complex real-world indoor environments. Also, the computational overhead of PSO optimization is higher than conventional LS methods, as the algorithm requires multiple iterations (75) to converge. Additionally, the effectiveness of the system is highly dependent on anchor placement, meaning that improper positioning of anchors may still lead to degraded accuracy.

An advanced TDoA-based UWB positioning system integrated with an EKF to enhance indoor localization accuracy is proposed by Trifunović et al., 2024 [40]. The EKF operates through two primary phases: the time update phase, where the system predicts the object's future position based on motion dynamics, and the measurement update phase,

where observed UWB signal data is employed to refine the position estimate. The experimental setup comprised four UWB anchors deployed in a rectangular configuration and a UWB tag serving as the tracked object. The results indicated that the EKF-based system achieved RMSE ranging from 0.40 m to 0.51 m. However, a notable limitation is the system's anchor placement dependency, as suboptimal anchor spatial distribution can degrade system performance. Furthermore, the study does not explicitly address NLOS mitigation, which represents a potential source of error in environments characterized by significant signal obstruction and multipath effects.

The work presented in [41] employs a ML-based NLOS detection method in conjunction with an adaptive robust extended Kalman filter (AREKF) for error compensation. The methodology comprises two primary components: support vector machine recursive feature elimination (SVM-RFE) for NLOS classification and AREKF for distance error mitigation. The SVM-RFE model extracts features from CIR data, selecting the most pertinent features through Pearson correlation coefficient (PCC) analysis. A hyperparameter tuning is performed utilizing GridSearchCV, and the AREKF algorithm applies adaptive corrections to refine ranging estimates. The system was validated using two datasets: a publicly available dataset from Ghent University and an internally collected dataset from an indoor office environment. The AREKF-SVM-RFE model achieved RMSE of 0.40 m in office settings and 0.50 m in hallway environments, with a mean error of 0.50 m. However, the positioning system's accuracy is contingent upon high-quality CIR feature extraction, implying that inadequate feature selection could degrade performance. Additionally, the computational complexity of AREKF and SVM-RFE exceeds that of conventional kalman filtering techniques. The system's efficacy is also influenced by anchor spatial distribution, as suboptimal configurations may introduce supplementary positioning errors.

III. PROBLE FORMALATION

UWB-based indoor positioning systems have emerged as a promising solution for high-precision localization in complex indoor environments. Whereas conventional methods such as Wi-Fi or Bluetooth-based positioning are highly susceptible to signal interference and multipath fading. UWB's fine time resolution enables centimeter-level accuracy, making them particularly valuable for applications in healthcare, industrial automation, robotics, and emergency response. However, despite these advantages, UWB-based IPS face persistent challenges which continue to limit its widespread and scalable deployment.

Key challenges in UWB positioning are the presence of MP interference, NLOS conditions, and the selection of reliable anchor nodes. NLOS conditions introduce significant biases in TOF measurements resulting in inaccurate distance estimations and consequently, leading to degraded localization accuracy. Conventional approaches typically rely on fixed anchor configurations, such as the geometric dilution of precision, and exhaustive geometric calculations, which limit their adaptability in cluttered and dynamic environments.

Moreover, existing anchor selection strategies either require a densely deployed network of anchors, increasing system complexity and cost, or employing computationally expensive optimization techniques, making them impractical for real-time applications. Binary LOS/NLOS identification methods also fail to account for real-time variations in channel conditions, limiting their effectiveness in rapidly changing
environments.

Furthermore, recent advancements in DL and ML have demonstrated strong potential for improving UWB positioning performance, particularly in NLOS mitigation and feature extraction. However, most existing models lack an adaptive anchor selection mechanism capable of dynamically prioritizing high-quality links while mitigating unreliable distance measurements. Many state-of-the-art methods suffer from limited generalization, as they do not incorporate adaptive mechanisms for anchor selection or real-time optimization under dynamic channel conditions.

Moreover, while neural networks have improved accuracy in signal classification, their computational complexity often hinders real-time feasibility.

Addressing these limitations requires an innovative and generalizable approach that effectively mitigates NLOS effects, performing intelligent anchor selection, and maintains high localization accuracy under diverse conditions. A scalable, low-complexity, and adaptive solution is essential to enable real-time deployment in dynamic UWB positioning systems. By integrating a lightweight deep learning framework with dynamic anchor ranking, sequential learning, and feature-driven bias mitigation, the proposed approach aims to bridge these gaps, enhancing both accuracy and computational efficiency for practical real-world applications.

IV. PROPOSED SYSTEM AND METHODOLOGY

The proposed LSB-DIP Net based AS algorithm aims to improve the accuracy of UWB indoor positioning across a range of environments, including challenging scenarios such as emergencies or unstructured environments. The algorithm facilitates sequential ranging between the tag and all anchors, providing the localization server with comprehensive channel information, such as received signal strength (RSL), first path signal level (FSL), preamble symbol count (PSC), and measured distances. Although this approach does not increase the update rate, it equips the localization server with extensive data for effective anchor selection.

The localization server employs a real-time LSB-DIP Net to identify the four anchors with the highest link quality (or three optimum anchors in the case of 2D positioning). The LSB-DIP Net uses a hierarchical ranking framework to

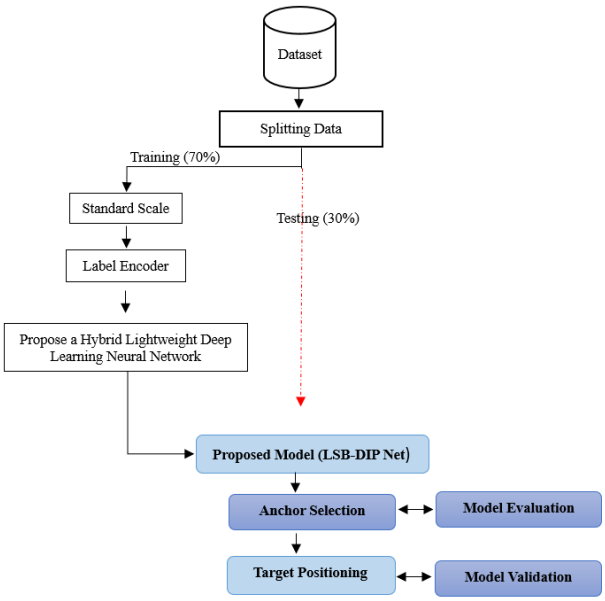


FIGURE 1. Proposed system architecture.

classify anchor channels as LOS, soft-NLOS, hard-NLOS, and MP, prioritizing LOS channels, followed by the others, respectively, to lessen distance bias. This hierarchical approach improves the accuracy of distance measurements for the selected anchors, ultimately enhancing positioning precision. The mobile station's position is then calculated using the LLS method.

The system architecture encompassing key components such as data preprocessing, the LSB-DIP Net model, positioning computations, and evaluation/validation phases are illustrated (Fig. 1). Algorithm 1 provides a step-by-step description of the proposed method.

A. DATA CHANNEL IMPULSE RESPONSE DATASETS OF UWB EXPERIMENTAL SETUP

The LSB-DIP Net methodology necessitates the availability of CIR data for each ranging measurement that is to be classified as LOS or NLOS. The experimental evaluations concentrated on UWB data measurement and used the Qorvo DW1000 module as our UWB hardware [42] for data creation. The Qorvo DW1000 can work well in high reflective RF conditions due to its resistance to rigorous multi-path conditions. This module is known to provide accurate and reliable UWB signal transmission and reception; hence, it was chosen in this work. The Qorvo DW1000 UWB module is a low-power, multichannel, single-chip CMOS radio transceiver that complies with the IEEE 802.15.4-2011 UWB standard. Utilizing TW-TOA, its range precision is ±10 cm. Also, it has six RF frequency bands spanning from 3.5 GHz to 6.5 GHz and a frequency bandwidth of 500–900 MHz Furthermore, it is energy efficient, can handle data transfers up to 6.8 Mbps, and can handle 110 kbps.

The initial dataset used in this research originates from our previous study [43], which involved collecting real measurement data under diverse propagation conditions, including LOS, MP, soft-NLOS, and hard-NLOS scenarios. Careful experimental design was implemented to gather data in various indoor settings, simulating a wide range of indoor environments. The data collection process introduced various movements and orientations to improve comprehensiveness and evaluate the adaptability of the UWB system in realistic scenarios.

Three supplementary datasets were also employed to evaluate the robustness and generalizability of the LSB-DIP Net mode. A dataset employed is based on the Bielefeld

Algorithm 1 Sequential steps of the proposed LSB-DIP Net framework for real-time UWB indoor positioning

Input:
- Set of anchor nodes A = {A1, A2, ..., An} with known positions (xi, yi, zi)
- Measured distances D = {d1, d2, ..., dn} from mobile station (MS)
- Received signal parameters (RSS, FSL, PSC)

Output:
- Estimated position of the mobile station $(\hat{x}, \hat{y}, \hat{z})$

Step 1: Initialize system
- Randomly install n anchor nodes in the indoor environment.
- Setup UWB environment and collect real-time CIR measurements (RSL, FSL, PSC and distance) from all anchors.

Step 2: Data Preprocessing
- Normalize input features features.
- Perform signal conditioning and noise reduction.
- Categorize channel conditions as {LOS, Soft-NLOS, Hard-NLOS, MP} for each anchor.

Step 3: Anchor Selection and Ranking
- Pass preprocessed features through LSB-DIP Net model.
- Rank anchors based on predicted link quality scores.
- Select the top four optimal anchors for positioning.

Step 4: Distance Bias Correction
- Mitigate distance bias from the selected anchors based on their classification.

Step 5: Position Estimation Using LLS
- Construct the system matrix A and vector b.
- Solve for $(\hat{x}, \hat{y}, \hat{z})$ using the least squares formulation: $(A^T A)^{-1} A^T b$

Step 6: Performance Evaluation
- Compute positioning accuracy metrics (RMSE, MSE, STD).
- Compare estimated position against ground truth.

Step 7: Real-time Adaptation
- Update system parameters dynamically for improved performance.
- Adjust model based on incoming data stream.

Return: Estimated mobile station position.

University 2020 research program [44] and contains three indoor scenarios classified as LOS, MP, and NLOS with different levels of data balance and sampling: unbalanced; under-sampled; and up-sampled. Moreover, an open-source UWB dataset originates from the IEEE Data port [22] and comprises three classes: LOS, soft-NLOS, and hard-NLOS. The last open-source dataset [45] was obtained from the eWINE project which measured across seven distinct indoor environments. These datasets provide a diverse and extensive foundation for evaluating the performance of the LSB-DIP Net methodology under various environmental conditions, ensuring its adaptability and reliability in real-world applications.

B. DATA PARTITIONING PREPROCESSING

The system follows a two-phase approach after partitioning the dataset into 70% for training and 30% for testing. This 70:30 data split ensures that the model is trained on a substantial portion of the dataset while reserving a distinct subset for rigorous evaluation of its generalization capabilities. Such a partitioning strategy is widely adopted in machine learning to balance model learning and performance assessment. The raw data undergoes a comprehensive preprocessing phase to prepare it for input into machine learning algorithms. Initially, categorical labels are transformed into numerical values to facilitate effective interpretation by the learning models. Subsequently, feature scaling is performed using standardization, where the data is rescaled to have a mean of zero and a standard deviation of one (unit variance). This normalization step is crucial for improving the convergence speed of optimization algorithms during training and enhancing the overall model performance by reducing sensitivity to feature scale variations.

The input data X is structured as a three-dimensional tensor with dimensions (N, T, F), where N represents the total measurements collected of samples, T denotes the sequence length expressed as the number of timesteps, and F indicates the number of features per timestep. This tensor representation allows the model to effectively capture temporal and spatial dependencies within the data, essential for robust prediction performance.

To ensure uniformity across features and enhance the predictive capability of the model, all input data undergo standardization during preprocessing. The transformation is mathematically defined as Eq. (1).

$$X' = \frac{X - \mu}{\sigma} \tag{1}$$

where μ is the mean of X, and σ denotes its standard deviation. This process standardizes the input data to have a mean of zero and unit variance. Standardization not only expedites the training process by improving convergence rates but also mitigates the risk of bias caused by feature scale disparities, thereby improving the overall reliability and accuracy of the model.

C. ANCHOR CHANNEL SELECTION AND RANKING MECHANISM

The proposed model introduces a new methodology for anchor channel selection and ranking, designed to enhance the accuracy and reliability of UWB indoor positioning systems. The framework addresses the inherent challenges of dynamic indoor environments, such as NLOS propagation and fluctuating signal conditions, by employing a hybrid sequential learning approach. The scalability and adaptability of the model make it a robust candidate for deployment across diverse indoor scenarios, providing reliable performance in real-time applications.

A key innovation of this framework lies in its adoption of a multi-optimal class anchor selection strategy, which classifies anchor channels into four optimal propagation conditions: LOS, MP, soft-NLOS, and hard-NLOS. This granular classification allows the model to account for temporal signal fluctuations and maintain stability in dynamic environments. By mitigating distance bias associated with selected anchor channels, the methodology ensures improved localization accuracy and reliability compared to binary LOS/NLOS classification techniques commonly employed in existing literature. This multi-class classification enables more robust and adaptive anchor selection, particularly in dynamic environments where channel conditions can exhibit rapid variations. Additionally, a distinguishing feature of the proposed model is its real-time evaluation framework, which assesses anchor channel quality and ranks anchors into multiple classes based on link quality. The ranking mechanism selects optimal anchors by analyzing an extensive dataset that reflects various indoor propagation conditions, including LOS, MP, soft-NLOS, and hard-NLOS. The used dataset serves as a comprehensive foundation for evaluating the model's performance. The framework not only identifies anchors with superior channel quality but also enhances positioning accuracy while maintaining computational efficiency.

The model employs a hybrid DL architecture that integrates multi-scale feature extraction, sequential learning, and advanced activation functions to capture temporal dependencies in dynamic indoor UWB positioning environments. These components enhance the system's ability to process complex propagation conditions and adapt to diverse indoor scenarios. Additionally, the architecture incorporates the conventional LLS method, combining traditional statistical techniques with advanced machine learning (ML) methods to optimize positioning performance.

The anchor channel selection and ranking mechanism proposed in this study offers a significant advancement in UWB indoor positioning systems by addressing critical limitations of traditional approaches. Its ability to perform multi-optimal class classification, coupled with a robust DL framework, ensures high localization accuracy and adaptability. These innovations position the model as a scalable and effective

solution for real-time applications in diverse and dynamic indoor environments.

D. TRILATERATION BASICS AND LEAST SQUARES APPLICATION IN POSITIONING SYSTEMS

Accurately determining the position of a mobile device requires precise distance estimates from known anchor points. These distances define circles (in 2D), or spheres (in 3D) centered at the anchor positions, with their radii representing the measured distances. Theoretically, the intersections of these circles or spheres correspond to the device's location.

Trilateration is a fundamental geometric approach for position estimation in UWB-based positioning systems. In a 2D environment, three anchor points are required to determine the location of an unknown target, whereas, in a 3D setup, four anchor points are needed. It often referred to the approach with interchangeably of true range multilateration [46], which is extended to 3D by requiring at least four spheres to determine the target's position accurately [46]. This method involves solving a set of nonlinear equations that describe the distances between the target and each anchor point. However, direct solutions to these nonlinear equations can be computationally intensive and impractical, especially in real-world scenarios with measurement noise and inaccuracies.

In a 3D positioning system, the goal is to determine the unknown coordinates (x, y, z) of a mobile device using known reference points, called anchors. Each anchor $\boldsymbol{A_i}$ is positioned at a known coordinate $(\boldsymbol{x_i}, \boldsymbol{y_i}, \boldsymbol{z_i})$. The Euclidean distance $\boldsymbol{d_i}$ between the target and each anchor is given by Eq. (2) [47], [48], [49].

$$(\boldsymbol{x_i} - \boldsymbol{x})^2 + (\boldsymbol{y_i} - \boldsymbol{y})^2 + (\boldsymbol{z_i} - \boldsymbol{z})^2 = \boldsymbol{d_i^2} \text{for } i = 1, 2, 3, 4 \tag{2}$$

Expanding and rearranging these equations yields a system of nonlinear equations, as in Eq. (3, 4, 5, 6).

$$\boldsymbol{x^2} - \boldsymbol{2x_1x} + \boldsymbol{x_1^2} + \boldsymbol{y^2} - \boldsymbol{2y_1y} + \boldsymbol{y_1^2} - \boldsymbol{2z_1z} + \boldsymbol{z_1^2} = \boldsymbol{d_1^2} \tag{3}$$
$$\boldsymbol{x^2} - \boldsymbol{2x_2x} + \boldsymbol{x_2^2} + \boldsymbol{y^2} - \boldsymbol{2y_2y} + \boldsymbol{y_2^2} - \boldsymbol{2z_2z} + \boldsymbol{z_2^2} = \boldsymbol{d_2^2} \tag{4}$$
$$\boldsymbol{x^2} - \boldsymbol{2x_3x} + \boldsymbol{x_3^2} + \boldsymbol{y^2} - \boldsymbol{2y_3y} + \boldsymbol{y_3^2} - \boldsymbol{2z_3z} + \boldsymbol{z_3^2} = \boldsymbol{d_3^2} \tag{5}$$
$$\boldsymbol{x^2} - \boldsymbol{2x_4x} + \boldsymbol{x_4^2} + \boldsymbol{y^2} - \boldsymbol{2y_4y} + \boldsymbol{y_4^2} - \boldsymbol{2z_4z} + \boldsymbol{z_4^2} = \boldsymbol{d_4^2} \tag{6}$$

Since these equations are nonlinear, solving for $\boldsymbol{x_i}, \boldsymbol{y_i}, \boldsymbol{z_i}$ directly requires numerical methods. However, we can linearize them to simplify computation, as shown in the following transformation. To solve for $\boldsymbol{x_i}, \boldsymbol{y_i}, \boldsymbol{z_i}$ the equations will be linearized by selecting one of them as a reference (such the equation for the first anchor, $i = 1$) and then subtracting it from the equations for the remaining anchors. This step eliminates quadratic terms and simplifies the expressions. The resulting set of linear equations is formulated as:

$$(\boldsymbol{x_i} - \boldsymbol{x_1})(x + \boldsymbol{x_1}) + (\boldsymbol{y_i} - \boldsymbol{y_1})(y + \boldsymbol{y_1}) + (\boldsymbol{z_i} - \boldsymbol{z_1})(z + \boldsymbol{z_1}) = \boldsymbol{d_i^2} - \boldsymbol{d_1^2} \tag{7}$$

For i = 2,3,4. The linearized system of equations can be expressed in matrix form as in in Eq. (8).

$$\boldsymbol{Ax} = \boldsymbol{b} \tag{8}$$

where,

$$A = \begin{bmatrix} x_2 - x_1 & y_2 - y_1 & Z_2 - Z_1 \\ x_3 - x_1 & y_3 - y_1 & Z_3 - Z_1 \\ x_4 - x_1 & y_4 - y_1 & Z_4 - Z_1 \end{bmatrix}, x = \begin{bmatrix} x_t \\ y_t \\ z_t \end{bmatrix},$$
$$\boldsymbol{b} = \frac{1}{2} \begin{bmatrix} \boldsymbol{d_2^2} - \boldsymbol{d_1^2} + \boldsymbol{x_2^2} - \boldsymbol{x_1^2} + \boldsymbol{y_2^2} - \boldsymbol{y_1^2} + \boldsymbol{z_2^2} - \boldsymbol{z_1^2} \\ \boldsymbol{d_3^2} - \boldsymbol{d_1^2} + \boldsymbol{x_3^2} - \boldsymbol{x_1^2} + \boldsymbol{y_3^2} - \boldsymbol{y_1^2} + \boldsymbol{z_3^2} - \boldsymbol{z_1^2} \\ \boldsymbol{d_4^2} - \boldsymbol{d_1^2} + \boldsymbol{x_4^2} - \boldsymbol{x_1^2} + \boldsymbol{y_4^2} - \boldsymbol{y_1^2} + \boldsymbol{z_4^2} - \boldsymbol{z_1^2} \end{bmatrix}$$

Solving x involves computing the least squares solution [50], [51], [52], [53]:

$$\boldsymbol{x} = (\boldsymbol{A^T A})^{-1} \boldsymbol{A^T b} \tag{9}$$

This formulation ensures computational efficiency while maintaining accuracy in real-world UWB-based positioning systems. In realistic indoor positioning scenarios, errors in distance measurements arise due to NLOS conditions, multipath effects, and signal interference. Consequently, the measured distances $\boldsymbol{d_i}$ may not correspond perfectly to the true distances. To mitigate these errors, the least squares estimation (LSE) method was employed. This approach minimizes the sum of squared errors between measured and estimated distances, ensuring a robust and reliable position estimate [51].

It should be noted that the elements of matrix A are taken from the reference coordinates of the anchor nodes, which are known to the system. Additionally, the elements of the vector b are also known, i.e. the measured distances (ranges) and the coordinates of the anchors. Therefore, the only unknown left is vector x, which is the location of the tag device. When more than the minimum number of anchor points are available (more than three in 2D or four in 3D), the problem becomes over-determined. The least squares approach offers computational efficiency for real-time processing with error minimization. The robustness of the least squares method ensures accurate position estimation even in noisy environments. By leveraging the redundancy of additional anchor points, this approach enhances positioning accuracy and reliability in real-world UWB-based systems.

E. PROPOSED LSB-DIP Net MODEL

DL, a primary subset of machine learning, is characterized by its architecture, which employs multiple layers of non-linear information processing arranged hierarchically. This approach is particularly advantageous for analyzing complex datasets [54]. A large amount of data is required for DL to extract features and train them effectively to identify the patterns and create meaningful weights. A traditional DL is developed with special hardware that has high specifications since the raw data used as input requires no pre-processing or feature extraction operations; hence, the training process involves a significant amount of time.

This paper introduces an adaptive, scalable, lightweight, hybrid deep neural network for assessing UWB anchor channel quality conditions in real-time for dynamic and complex indoor positioning scenarios. The proposed method enables the selection of optimal anchors for indoor positioning systems across diverse environments, including challenging scenarios. The approach ranks anchors based on the degree of obstruction, considering a continuum of channel conditions such as LOS, soft-NLOS, hard-NLOS, and MP channels. The proposed LSB-DIP Net is a one-dimensional neural network adapted from the traditional two-dimensional neural network. This method, owing to its simplicity in computations and speed, can be executed in cost-effective and simple environments, making it useful for real-time practical applications.

1) CONCEPTUAL OVERVIEW OF LSB-DIP Net

The LSB-DIP Net model is designed to enhance UWB-based indoor positioning by integrating adaptive anchor selection and multi-class classification of channel conditions, thereby improving localization accuracy. Unlike conventional methods that rely on fixed anchor configurations, the proposed model dynamically evaluates and selects the most reliable anchors based on real-time signal characteristics. This adaptability ensures robust and precise localization, even in complex and dynamically changing indoor environments. At its core, LSB-DIP Net incorporates DL-based feature extraction, sequential modeling, and classification layers to efficiently process UWB signals. The model follows a structured data processing pipeline, where raw UWB measurements are transformed through multiple stages to extract meaningful features, classify anchor channel conditions, and ultimately rank and select the most reliable anchors for position estimation.

The feature extraction process is performed using one-dimensional convolutional layers (Conv1D), which are responsible for identifying spatial patterns in UWB signals. These layers apply multiple filters to detect variations in signal amplitude, distortion, and noise, extracting relevant features that distinguish between LOS, soft-NLOS, hard-NLOS, and MP conditions. Additionally, to account for time-dependent variations in signal propagation, the model incorporates LSTM layers. These layers are particularly beneficial in dynamic indoor environments, where signal conditions fluctuate due to obstacles, movement, and multipath effects.This enhances the robustness of the classification process, ensuring that anchor selection remains adaptive to real-time environmental changes. Furthermore, to improve computational efficiency, MaxPooling layers are employed for dimensionality reduction while preserving crucial signal characteristics. These layers selectively retain the most important features by downsampling the feature maps, reducing redundancy and mitigating the impact of noise. Finally, the classification and decision-making process is performed through fully connected dense layers followed by a Softmax activation function. These layers aggregate the extracted spatial and temporal features to assign probability scores to each propagation condition. The Softmax function ensures that the classification outputs form a probability distribution, allowing the system to make confident predictions regarding the quality of each anchor's signal link. This classification enables the dynamic ranking and selection of optimal anchors, ensuring that only the most reliable anchors contribute to the final position estimation process.

This layered hierarchical structure enables LSB-DIP Net to effectively process sequential signal data, classify distinct propagation conditions, and dynamically rank anchors, ensuring optimal positioning performance.

2) DETAILED ARCHITECTURE OF LSB-DIP Net

The LSB-DIP Net model consists of several processing stages, each serving a distinct role in transforming raw UWB data into a robust positioning solution. This application will feed each input through repeated convolutional layers with kernels, pooling, LeakyRelu, LSTM, and fully connected layers to classify the UWB propagation channel conditions with very high accuracy.

The seven one-dimensional convolutional layers belonging to the LSB-DIP Net model having 24 layers are tasked with the extraction of prominent features. Each convolution operation applies a set of filters (W) over the input (X) to generate feature maps, as described by Eq. (10).

$$Z_{i,j} = \sum_{k=1}^{K} W_k . X_{i+k,j} + b \tag{10}$$

where $Z_{i,j}$ represent the output feature map at position (i, j), W_k denotes the weights of the filter, $X_{i+k,j}$ refers to the corresponding slice of the input data, b represents the bias term added to the result, and K Indicates the size of the convolutional kernel. This process extracts meaningful features from the input data by applying learnable filters across temporal dimensions, enabling the model to capture patterns relevant to the task. The output shape could be determined by the padding and stride parameters as shown in Eq. (11).

$$L_{OUT} = \frac{L_{IN} - k + 2.D}{S} + 1 \tag{11}$$

where L_{OUT}, L_{IN}, D and S represent the Length of the output sequence, Length of the input sequence, Padding size, and Stride respectively.

Six max-pooling layers to enhance the model's translational invariance, ensuring that minor spatial shifts in the input do not significantly alter the network's output and extract the most salient features from the preceding convolutional layer, effectively reducing the dimensionality of the representation while preserving the most critical information. The application of max-pooling layers serves to downsample the feature representations produced by convolutional operations. This dimensionality reduction facilitates a decrease in computational complexity and enhances the model's robustness to small spatial variations in the input data. Dimensionality reduction is achieved through max pooling, where the output feature map is computed as in Eq. (12).

$$Z_{i,j} = max(X_{i:i+p,j:j+p}) \tag{12}$$

Where p is the pooling window size.

Seven Leaky Rectified Linear Unit (LeakyReLU) activation layers were included to introduce non-linearity that mitigates the "dying ReLU" problem, a phenomenon where ReLU neurons become inactive due to zero gradients for negative inputs. Unlike the standard ReLU, which outputs zero for x<0, LeakyReLU introduces a small, non-zero slope for negative values, expressed Eq. (13).

$$f(x) = \begin{cases} x & if\ x < 0 \\ x & if\ x \geq 0 \end{cases} \tag{13}$$

where x is the input to the activation function, and δ is a small positive constant (such as 0.3). By allowing a non-zero gradient for negative inputs, LeakyReLU prevents neurons from becoming permanently inactive and ensures continued learning during the training process.

Two LSTM layers were incorporated into the architecture to effectively capture the complex temporal dynamics within the sequential features extracted by the preceding convolutional layers. LSTM networks constitute a specialized class of recurrent neural networks (RNNs) designed to address the challenge of capturing long-term dependencies in sequential data, a critical aspect for accurate modeling of time-series phenomena [55]. This capability is achieved through the incorporation of a memory cell, complemented by a triad of gating mechanisms: the input, forget, and output gates. These gates function selectively, determining which information should be retained within the cell state and which should be discarded based on the current input, the preceding output, and the current cell state. In the context of a given sequence $x = (x_1, x_2, \ldots, x_n)$, an initial state is determined through a forward calculation process, as in Eq. (14, 15, 16, 17, 18).

$$i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + W_{ci}c_{t-1} + b_i) \tag{14}$$
$$f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + W_{cf}c_{t-1} + b_f) \tag{15}$$
$$o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + W_{co}c_t + b_o) \tag{16}$$
$$c_t = f_t c_{t-1} + i_t tan\ h(W_{xc}h_{t-1} + W_c c_{t-1} + b_c) \tag{17}$$
$$h_t = o_t tan\ h(c_t) \tag{18}$$

Within this framework, *i, f,* and *o* denote the input, forget, and output gates, respectively. The variable c represents the cell state, while h signifies the hidden layer sequence, which can also be interpreted as the output sequence corresponding to the current input. W and b represent the weight matrices and bias terms, respectively, with (tanh) hyperbolic tangent and (σ) sigmoid serving as the activation function. Following the forward pass, the LSTM network undergoes backpropagation training. This process commences with the backward calculation of the error associated with each cell. Subsequently, the gradient of each weight is computed based on the calculated errors. Finally, the model weights are updated according to the computed gradients through the application of an appropriate optimization algorithm. Also, a flatten layer is employed to transform the multidimensional output generated by preceding layers into a one-dimensional vector, ensuring compatibility with subsequent layers that expect a one-dimensional input. The final output Dense Layer with SoftMax Activation Function is used to categorize the features derived from the previous layers. The SoftMax function converts these features into values between 0 and 1, enabling their interpretation as probabilities. Inputs that are small or negative are transformed into low probabilities, while larger inputs are converted into higher probabilities, yet they consistently remain within the 0 to 1 range. A final dense, fully connected layer makes predictions based on learned features, as in Eq. (19).

$$P(y_i) = \frac{e^{z_i}}{\sum_j e^{z_j}} \tag{19}$$

where the z_i is the logit for class i, and $P(y_i)$ is the probability of the true label.

The model is trained using the Adam optimizer with a learning rate of 10^{-3}, a batch size of 1024, and is executed for 100 epochs. The layers of the LSB-DIP Net model with more information are listed (Table 1). The model's architecture consists of 24 layers, including nine one-dimensional convolutional layers, eight max-pooling layers, five dense layers, and one flattening layer (Fig. 2).

3) ADAPTIVE ANCHOR SELECTION AND POSITION ESTIMATION

Following the classification of anchor channels, the system employs a hierarchical ranking strategy to identify and select the most reliable anchors for positioning. LOS anchors are assigned the highest priority, followed by soft-NLOS, hard-NLOS, and MP conditions, respectively. This prioritization is based on classification accuracy, ensuring effective mitigation of bias errors introduced by NLOS conditions.

The dynamic anchor selection process optimizes the identification of high-quality anchor links, thereby enhancing positioning accuracy and stability. Once the optimal anchor set is determined, the LLS method is applied to compute the mobile station's position using the selected high-reliability anchors. This approach ensures low positioning error and robust localization performance, even in complex indoor environments.

F. KEY ADVANTAGES OF THE PROPOSED LSB-DIP Net FRAMEWORK

The proposed system introduces a novel framework for UWB indoor positioning, using powerful temporal processing techniques to effectively address the inherent challenges posed by complex indoor environments, specifically those characterized by NLOS propagation. This innovative approach offers several key advantages, including:

1) The integration of multi-scale feature extraction employing filters of varying sizes significantly enhances the signal processing capabilities of UWB indoor positioning systems. This approach enables the model to effectively capture both localized details (small

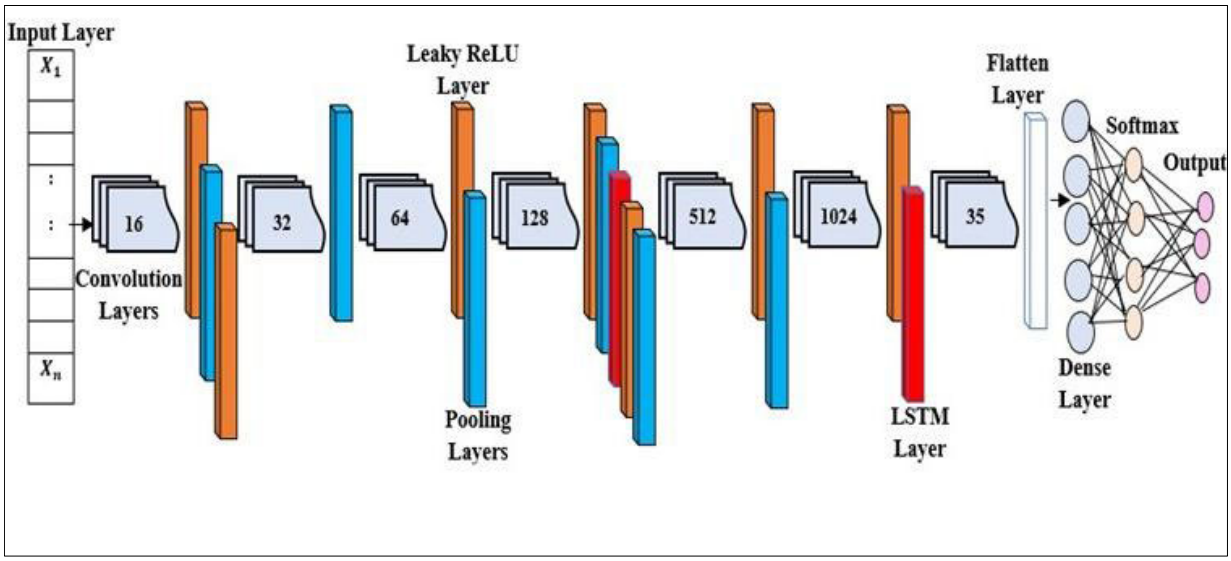


FIGURE 2. The proposed LSB-DIP Net model.

variations) and broader global patterns (signal trend and dynamic changes), leading to a more robust and comprehensive understanding of signal characteristics. Additionally, it improved resilience to environmental variations, including those arising from multi-class anchor propagation channels. Furthermore, this approach improves resilience to environmental variations and ensures robust signal interpretation across multiple levels of abstraction.

2) Temporal Context Preservation in UWB indoor positioning systems enhances positioning accuracy by utilizing sequential signal patterns, enabling robust predictions in dynamic and noisy environments. It reduces susceptibility to sudden signal fluctuations by considering broader temporal trends, thereby improving reliability. Additionally, this approach supports generalization across diverse environmental conditions and ensures adaptability to real-time data streams, maintaining consistent and accurate performance.
3) The innovative application of categorical cross-entropy loss in this UWB-based indoor positioning system effectively optimizes multi-class anchor channel quality classification. By aligning predicted probabilities with true position labels and penalizing incorrect predictions with higher penalties for low probabilities of the true class, this loss function drives the model towards confident and precise decisions, significantly enhancing the system's robustness, adaptability, and accuracy in real-time indoor positioning tasks.
4) The model exhibits strong adaptability to dynamic indoor environments, where positions and signal conditions fluctuate rapidly, due to the inherent capabilities of LSTMs in handling temporal signal variations. This adaptability ensures robust real-time performance. Furthermore, the integration of lightweight 1D convolutional layers with efficient pooling strategies significantly reduces computational complexity, facilitating faster inference and ensuring low latency, a critical requirement for real-time deployment. The implementation of batch processing and the utilization of the Adam optimizer for efficient optimization expedite convergence, further enhancing the model's suitability for real-time applications.
5) The inherent modularity of the model facilitates seamless deployment across diverse indoor environments, minimizing the need for extensive re-engineering efforts. This modular architecture exhibits inherent scalability and adaptability, allowing for the flexible adjustment of hyperparameters such as the number of filters, kernel sizes, and LSTM units. This adaptability enables the model to effectively accommodate the unique requirements and complexities inherent in various indoor environments and signal characteristics.
6) The proposed UWB-based indoor positioning system, classification-based positioning simplifies the task of localizing objects by utilizing the proposed techniques to classify multiple anchor channel quality rather than traditional classification-based localization, where each class represents a predefined specific propagation channel. This approach improves the system's accuracy, robustness to noise, and adaptability to dynamic environments, making it suitable for real-time applications.
7) The model architecture incorporates MaxPooling1D layers, effectively down sampling the feature maps while preserving salient information and mitigating the risk of overfitting. Furthermore, the strategic implementation of LeakyReLU activation functions, carefully

TABLE 1. Layers of the proposed LSB-DIP Net model.

No.	Layer	Parameter	Output Shape
1	conv1d-1 (Conv1D)	64	(None, 2, 16)
2	leaky_re_lu_1 (LeakyReLU)	0	(None, 2, 16)
3	max_pooling1d-1 (MaxPooling1	0	(None, 1, 16)
4	leaky_re_lu_2 (LeakyReLU)	0	(None, 1, 16)
5	conv1d-2 (Conv1D)	1568	(None, 1, 32)
6	max_pooling1d-2 (MaxPooling1	0	(None, 1, 32)
7	conv1d-3 (Conv1D)	6208	(None, 1, 64)
8	leaky_re_lu_3 (LeakyReLU)	0	(None, 1, 64)
9	max_pooling1d-3 (MaxPooling1	0	(None, 1, 64)
10	conv1d-4 (Conv1D)	24704	(None, 1, 128)
11	leaky_re_lu_4 (LeakyReLU)	0	(None, 1, 128)
12	max_pooling1d-4 (MaxPooling1	0	(None, 1, 128)
13	lstm_1 (LSTM)	20608	(None, 1, 32)
14	leaky_re_lu_5 (LeakyReLU)	0	(None, 1, 32)
15	max_pooling1d-5 (MaxPooling1	0	(None, 1, 32)
16	conv1d-5 (Conv1D)	49664	(None, 1, 512)
17	leaky_re_lu_6 (LeakyReLU)	0	(None, 1, 512)
18	max_pooling1d-6 (MaxPooling1	0	(None, 1, 512)
19	conv1d-6 (Conv1D)	1573888	(None, 1, 1024)
20	leaky_re_lu_7 (LeakyReLU)	0	(None, 1, 1024)
21	lstm_2 (LSTM)	135296	(None, 1, 32)
22	conv1d-7 (Conv1D)	3395	(None, 1, 35)
23	flatten_1 (Flatten)	0	(None, 35)
24	dense_1 (Dense)	144	(None, 4)
		Total params: 1,815,539	

selected to enhance model stability, provides an effective regularization strategy.

G. THE LSB-DIP Net MODEL LAYERS CONFIGURATION

Compared to traditional DL models that extract features from local patterns, the LSB-DIP Net model combines spatial and temporal learning to capture relationships across sequential data. By optimizing feature extraction and sequence modeling, this architecture reduces reliance on large datasets, mitigates overfitting through hybrid learning, and efficiently handles complex patterns in dynamic environments. The next paragraph sheds light on all 24 layers of this unique architecture:

1) Convolution 1D Layer (16 Filters): This layer serves as the first feature extraction stage, utilizing 16 filters, each with a kernel size of 3. It processes triplets of input elements with a stride of 1 and employs a valid padding scheme, meaning no padding is added, ensuring the input dimensions remain unchanged after processing.
2) LeakyReLU Activation Layer: Acting as a feature normalization layer, this activation introduces non-linearity while maintaining the tensor dimensions. With a slope of 0.3 for negative inputs, it ensures that neurons remain active, even for negative values.
3) MaxPooling Layer: This feature selection layer performs downsampling by selecting the highest value in each pooling window. It operates with a pool size of 2, a stride of 2, and a same padding scheme, reducing the dimensionality of the feature map without overlapping pooling windows.
4) LeakyReLU Activation Layer: Similar to the previous activation layer, it introduces non-linearity while maintaining the same slope of 0.3 for negative inputs, ensuring smooth gradient flow.
5) Convolution 1D Layer (32 Filters): This layer extracts features using 32 filters, each with a kernel size of 3, a stride of 1, and a same padding scheme. The convolution operation generates a detailed feature map for subsequent layers.
6) MaxPooling Layer: A repeated feature selection stage that downsamples the input, this layer has a pool size of 2, a stride of 2, and a same padding scheme, effectively reducing dimensionality.
7) Convolution 1D Layer (64 Filters): Building on the prior convolution, this layer employs 64 filters, each with a kernel size of 3, a stride of 1, and a same padding scheme, further enhancing feature extraction.
8) LeakyReLU Activation Layer: As with earlier instances, this activation layer introduces non-linearity, allowing small gradients to pass through negative inputs, with the same slope value of 0.3.
9) MaxPooling Layer: Another feature selection layer, this one continues to downsample input data with a pool size of 2, a stride of 2, and a same padding scheme.
10) Convolution 1D Layer (128 Filters): This layer utilizes 128 filters with the same configuration as earlier convolution layers, amplifying feature extraction and reinforcing the detection of relevant patterns.
11) LeakyReLU Activation Layer: Again, this activation layer introduces non-linearity with a slope of 0.3 for negative inputs, ensuring smooth training dynamics.
12) MaxPooling Layer: Serving as a dimensionality reduction step, this layer applies the same pool size, stride, and padding as previous pooling layers, maintaining consistency in feature selection.
13) LSTM Layer (32 Units): This layer captures temporal dependencies in sequential data with 32 units, each maintaining its internal cell state to preserve long-term information. By setting "return_sequences = True," it outputs a sequence of hidden states for each time step,

enabling compatibility with subsequent layers requiring sequential input.

14) LeakyReLU Activation Layer: As in previous occurrences, this layer introduces non-linearity and smooth gradients for negative inputs with a slope of 0.3.
15) MaxPooling Layer: Continuing the dimensionality reduction process, this layer maintains the same pool size, stride, and padding scheme as earlier pooling layers.
16) Convolution 1D Layer (512 Filters): With 512 filters, each with a kernel size of 3, a stride of 1, and a same padding scheme, this layer significantly boosts feature extraction capabilities.
17) LeakyReLU Activation Layer: Consistent with earlier instances, this layer ensures that small gradients flow for negative inputs, using a slope of 0.3 to enhance pattern learning.
18) MaxPooling Layer: In this step, the pool size is 2, the stride is also 2, and a same padding scheme is applied, further reducing the dimensionality of the feature maps.
19) Convolution 1D Layer (1024 Filters): This layer maximizes feature extraction using 1024 filters, maintaining the kernel size, stride, and padding configuration consistent with previous convolution layers.
20) LeakyReLU Activation Layer: This layer, as before, applies a non-linear activation function, allowing gradients to flow for negative inputs with a slope of 0.3.
21) LSTM Layer (32 Units): A second LSTM layer identical in configuration to the first, this layer reinforces the model's ability to capture and process temporal dependencies within the sequential data.
22) Convolution 1D Layer (35 Filters, Linear Activation): Unlike prior convolution layers, this one employs a linear activation function, performing the convolution operation without introducing non-linear transformations. With 35 filters and the same kernel size, stride, and padding configuration, this layer primarily serves as a feature transformation mechanism.
23) Flatten Layer: This layer reshapes the multi-dimensional feature maps into a one-dimensional vector, preparing the data for the dense layer.
24) Dense Layer (Softmax Activation): The final classification layer outputs a probability distribution across target classes. Using the softmax function, it normalizes the output values, ensuring they sum to 1, providing probabilistic predictions for each class.

V. EXPERIMENTAL EVALUATION AND VALIDATION

The evaluation and validation of the proposed system were conducted in a structured manner, as outlined in Algorithm 1 and divided into two distinct phases. The first phase assessed the real-time anchor selection method, focusing on its ability to identify anchors with high-quality channels while minimizing ranging errors across various environmental conditions. This evaluation aimed to demonstrate the method's robustness and adaptability to diverse scenarios. The second phase validates the system's performance in real-time positioning applications. Specifically, the system's error metrics and positioning accuracy were analyzed for an unknown target moving along different trajectories in challenging indoor environments. This two-phase approach ensured a comprehensive assessment of the proposed system under simulated and practical conditions.

To ensure a rigorous evaluation, a wide range of performance metrics was employed. Classification metrics, including accuracy, precision, recall, and F-measure, were utilized to measure the system's effectiveness in channel condition classification. Error metrics, such as RMSE, Maximum Absolute Error (Max Error), Standard Deviation of Error (σ), and the Cumulative Distribution Function (CDF) of Error, quantified the accuracy of the positioning system. Computational efficiency was also evaluated by analyzing the processing time required for real-time operation.

The implementation of the proposed system was carried out on a high-performance computing platform. The hardware configuration included a Lenovo laptop equipped with an Intel Core i7-12700HQ processor (12th generation, 2.7 GHz, 8 MB cache), 16 GB of RAM, and an NVIDIA GeForce RTX 3050 Ti graphics processor. The system development and testing were performed using Python and MATLAB R2022a, leveraging their advanced computational and simulation capabilities to implement the LSB-DIP Net-based framework effectively. The Lenovo laptop was configured to function as the localization server within the experimental setup. This configuration involved establishing a connection between the laptop and a single anchor node to enable the reception of UWB measurements. The localization server is responsible for real-time processing and analysis of these measurements. This process was executed through the LSB-DIP Net model, which facilitated adaptive anchor selection and enabled the accurate computation of the mobile station's position.

The proposed system was evaluated through real-world experiments involving both a human participant and a mobile robot, ensuring a comprehensive assessment under different motion conditions. As described in our prior work [43], a human participant carried the UWB device while traversing predefined paths, introducing natural movement variations, including changes in walking speed, orientation shifts, and hand placement. This setup effectively simulated realistic wearable UWB positioning applications, such as indoor navigation and asset tracking, in both static and dynamic environments. The variability introduced by human motion aimed to replicate practical real-world positioning scenarios, where factors such as arm swings, random pauses, and orientation deviations influence localization accuracy. To further validate the system under controlled motion conditions, a mobile robot equipped with a Qorvo DW1000 UWB transceiver was employed. The robot was remotely controlled to follow predefined trajectories, ensuring consistent and repeatable motion paths for systematic evaluation.

TABLE 2. Ranging error across various channels.

Actual Distance (m)	LOS		Soft-NLOS		Hard-NLOS		MP	
	Measured Distance (m)	Bias (m)	Measured Distance (m)	Bias (m)	Measured Distance (m)	Bias (m)	Measured Distance (m)	Bias (m)
1	1.03	0.03	1.41	0.41	..	..	..	..
2	2.00	0.00	2.37	0.37	..	..	2.33	0.33
3	3.03	0.03	3.34	0.34	4.17	1.17	3.47	0.47
4	4.07	0.07	4.35	0.35	5.22	1.22	4.46	0.46
5	5.07	0.07	5.34	0.34	6.02	1.02	5.31	0.31
6	6.06	0.06	6.34	0.34	7.13	1.13	6.55	0.55
7	7.05	0.05	7.34	0.34	8.02	1.02	7.58	0.58
8	8.03	0.03	8.32	0.32	9.02	1.02	8.27	0.27
9	9.08	0.08	9.29	0.29	10.23	1.23	9.61	0.61
10	10.02	0.02	10.36	0.36	11.19	1.19	10.32	0.32
11	11.02	0.02	11.29	0.29	12.21	1.21	11.53	0.53
12	12.05	0.05	12.34	0.34	13.21	1.21	12.39	0.39
13	13.05	0.05	13.25	0.25	14.20	1.20	13.23	0.23
14	14.04	0.04	14.41	0.41	15.22	1.22	14.40	0.40
15	15.06	0.06	15.27	0.27	16.21	1.21	15.38	0.38
16	16.09	0.09	16.34	0.34	17.21	1.21	16.40	0.40

Empirical evaluation of the LSB-DIP Net algorithm was conducted using four distinct datasets. The first dataset, Ds1, was derived from the Bielefeld University 2020 research program [43] and comprised three indoor scenarios categorized as LOS, MP, and NLOS. This dataset was further divided into subsets to address different levels of data balance: unbalanced (Ds1-1), under-sampled (Ds1-2), and up-sampled (Ds1-3). The second dataset, Ds2, was obtained from the IEEE Data port [22] and included three classes: LOS, soft-NLOS, and hard-NLOS. The third dataset, Ds3, was generated through experimental processes conducted in prior work [43] and contained four propagation channels: LOS, MP, soft-NLOS, and hard-NLOS. The last open-source dataset (Ds4) [45] was obtained from the eWINE project which measured across seven distinct indoor environments. These datasets provided a diverse range of scenarios to evaluate the model's generalizability and performance across varying conditions.

To complement the simulations, a real-world experiment was conducted in a complex indoor environment. The test area, a 20m $\times$ 20m rectangular space, featured 30 cm thick concrete walls and various office furniture to replicate realistic conditions. Six anchor nodes (A1 to A6) were deployed at fixed coordinates: A1$(-3, 8.5)$, A2$(7, 9)$, A3$(9, -1)$, A4$(-4, -1)$, A5$(11, 4)$, and A6$(0, -6.5)$. Fig. 3 illustrates the environment and the anchor node deployment. A series of 90-meter trajectories were designed to simulate the movement of a target mobile station equipped with a Qorvo DW1000 UWB device [42]. This device facilitated communication with the anchor nodes, collecting eight distance measurements per second. Operating on UWB channel 2 with 1024 preamble symbols and a 64 MHz pulse repetition frequency (PRF), the module ensured a stable and reliable communication link.

The real-world test environment was equipped with a high-precision distance measurement system (GLM 100-25 C Professional) to ensure accurate ground truth data and reliable trajectory simulation. Each measurement included RSS, FSL, PSC, and distance. These data served as inputs to the proposed LSB-DIP Net algorithm for channel condition classification.

The LSB-DIP Net model classified the channel conditions of each anchor link as LOS, soft-NLOS, hard-NLOS, or MP. Real-time evaluation of channel quality enabled the selection of the four best-performing anchors for precise localization. To further enhance accuracy, distance bias mitigation techniques were applied to the selected anchors. The final position of the mobile station was estimated using the LLS method, leveraging the optimized anchor set.

A detailed analysis of bias in measured distances across the 1–16 m range, highlighting the impact of varying propagation conditions, such as LOS, MP, soft-NLOS, and hard-NLOS, on system performance is provided (Table 2). These findings underscore the system's adaptability and reliability, demonstrating its effectiveness in mitigating ranging errors and achieving precise positioning in complex environments.

The experimental results emphasize the applicability of UWB technology in challenging indoor environments, such as military operations and emergency responses, within unknown environments [12], [13], [56] where precise

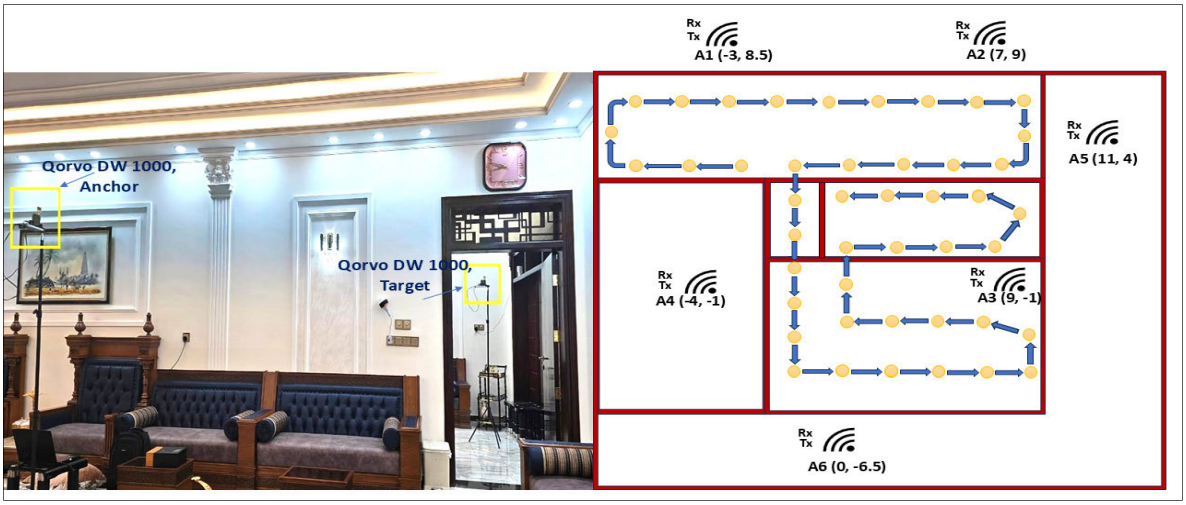


FIGURE 3. Real-time environment with a schematic diagram of robot trajectory.

positioning is critical. By integrating DL techniques for link quality estimation and error mitigation, the system addresses key challenges in NLOS environments, offering a robust solution for real-time monitoring and control in operational activities.

VI. RESULTS AND DISCUSSION

As discussed in the existing literature on UWB indoor positioning systems highlights several limitations, such as poor generalizability, challenges in lessening NLOS and multipath effects, high computational demands, and sensitivity to environmental factors. This work addresses these challenges by developing a precise, efficient, and robust indoor positioning system capable of operating in diverse and unpredictable environments. The main contributions include: (i) A novel LSB-DIP Net-based AS algorithm for improving UWB indoor positioning accuracy in diverse environments, including emergency scenarios. (ii) A real-time evaluation mechanism to select optimal anchors by assessing channel conditions (LOS, soft-NLOS, hard-NLOS, and multipath). (iii) Rigorous experimental validation demonstrates noticeable localization accuracy improvements, with a maximum positioning error of 0.17 m across diverse scenarios.

The adoption of a multi-class anchor classification approach is motivated by its ability to more accurately characterize varying levels of signal obstruction and degradation in dynamic indoor environments. Unlike binary classification, which simply distinguishes between LOS and NLOS, our method offers granular differentiation among LOS, soft-NLOS, hard-NLOS, and MP conditions. This refinement enables more precise bias correction and optimal anchor selection, ultimately enhancing localization accuracy. Empirical findings, as presented in Table 4, substantiate the need for this multi-class approach. Our results indicate that soft-NLOS conditions exhibit a moderate-ranging bias of 0.29 to 0.41 m while still preserving usable signal integrity. In contrast, hard-NLOS conditions introduce a significantly higher bias (greater than 1.0 m), which, if not correctly identified, leads to severe localization inaccuracies. Additionally, MP conditions induce erratic distance fluctuations, often misclassified under binary schemes as either LOS or NLOS, resulting in suboptimal bias correction and anchor selection. One of the primary advantages of multi-class classification is its ability to leverage both soft and hard NLOS anchors strategically. By selectively integrating soft and hard NLOS anchors with appropriate weighting rather than rejecting them outright, our approach improves localization precision without compromising robustness. Furthermore, distinguishing between hard-NLOS and MP conditions enables the implementation of adaptive error mitigation strategies. These include adjusting the weighting of high-quality anchors, selectively excluding only the most degraded signals, and dynamically adapting anchor selection criteria based on environmental conditions. The findings indicate that the absence of a multi-class anchor classification strategy, where NLOS conditions are treated as distinct categories, may adversely affect the performance of existing methods. This limitation can lead to reduced accuracy in range measurements, identification outcomes, and ultimately, the precision of target localization, as demonstrated in [25] and [26]. Although some studies, such as [29] and [31] report high accuracy under controlled conditions, their performance tends to degrade in more complex or dynamic environments. In contrast, the proposed multi-class approach outperformed the binary model across all evaluation metrics. Specifically, the positioning error was reduced from ± 50 cm to 15.86 cm, reflecting a 68.4% improvement, while the NLOS identification accuracy increased to 99%. These results highlight the enhanced adaptability of the multi-class framework and underscore its effectiveness in delivering accurate, robust, and adaptive performance across a wide range of indoor scenarios. The LSB-DIP Net, by incorporating this refined

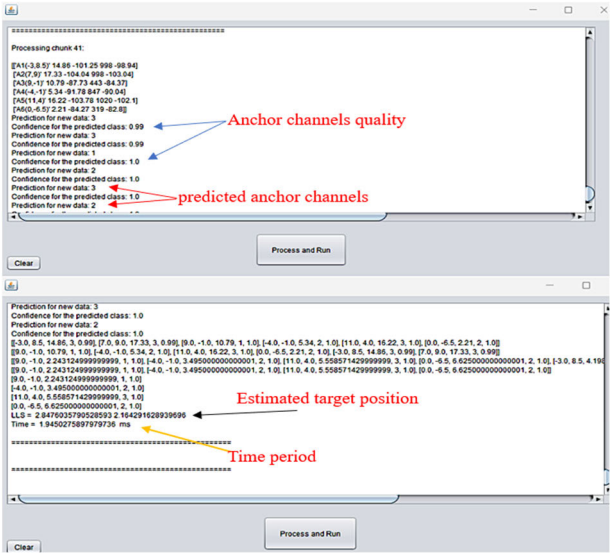


FIGURE 4. The graphical user interface interface of the proposed LSB-DIP Net system, showing real-time positioning estimation.

classification framework, significantly outperforms traditional binary approaches in terms of accuracy, robustness, and adaptability.

The real-time control process of our proposed LSB-DIP Net as displayed by the main server is illustrated (Fig. 4). The figure depicts a series of discrete stages (chunks), each chunk representing a real-time streaming processing step. These stages encompass anchor channel prediction and online evaluation, optimal anchor selection, error mitigation, priority assignment based on obstacle severity, and final position estimation with high accuracy, along with the corresponding processing time. The LSB-DIP Net demonstrates an average end-to-end processing latency of 1.945 ms per positioning cycle, from anchor selection to position estimation, thereby indicating its capacity for real-time performance in dynamic environments.

The initial data entry within each processing chunk, denoted by the yellow arrow, comprises the coordinates of the first known anchor, followed by the estimated distances and extracted features, with subsequent lines containing data for the remaining anchors. For each processed data chunk, a predicted anchor channel (indicated by the red arrow) and its associated confidence score (indicated by the blue arrow) are generated. The consistently high confidence scores observed for the predicted anchor channels, typically at or near 1.0, demonstrate a high degree of certainty in the model's predictions. This level of confidence suggests robust classification capabilities within the system. Furthermore, the predicted anchor channel designations (1, 2, 3, and 4) correspond to anchors located within LOS, soft-NLOS, hard-NLOS, and MP propagation channels, respectively.

The experimental evaluation utilized both real-world and open-source datasets (DS1–DS4), providing diverse and

TABLE 3. Summary of datasets and characteristics.

Dataset	Environment	Propagation Conditions	Total Samples
DS1-1	Small rooms, halls, corridors	LOS, MP, NLOS	188,791
DS1-2	Small rooms, halls, corridors	LOS, MP, NLOS	335,539
DS1-3	Small rooms, halls, corridors	LOS, MP, NLOS	273,041
DS2	Corridors, offices	LOS, soft-NLOS, hard-NLOS	30,732
DS3	Residential rooms, hall, corridors	LOS, MP, soft-NLOS, hard-NLOS	590,940
DS4	Various indoor	LOS, NLOS	42,000

TABLE 4. Comparison of results for LSB-DIP Net technique using different datasets.

Anchor channel	Accuracy %	Precision %	Recall %	F1-score %	Loss Function	No. Epoch	Time/epoch
LOS	100	100	100	100	0.365	250	4s
Soft-NLOS	99.0	99.0	99.0	99.0	0.365	250	4s
Hard-NLOS	98.0	98.0	98.0	98.0	0.365	250	4s
MP	99.0	99.0	99.0	99.0	0.365	250	4s

extensive test scenarios. Table 3 summarizes the datasets used, covering various indoor environments such as residential rooms, corridors, halls, and office spaces. Propagation conditions include LOS, MP, soft-NLOS, and hard-NLOS, ensuring a comprehensive analysis of the system's performance.

The LSB-DIP Net scheme was evaluated using key metrics, including accuracy, precision, recall, F1-score, and computational computational efficiency. Table 4 summarizes the classification performance of the model across all datasets.

The results demonstrate that the LSB-DIP Net scheme consistently achieves exceptional accuracy, with values exceeding 98% across all datasets. Precision, recall, and F1-scores were near-perfect in most cases, particularly under challenging conditions such as soft-NLOS and hard-NLOS. This indicates the model's capability to minimize false positives while effectively identifying true positives, ensuring reliable performance under diverse scenarios. This efficiency emphasizes the model's suitability for applications where both accuracy and speed are critical.

The LSB-DIP Net underwent a comprehensive regularization process to rigorously address the concern risk of overfitting. Initially, the model was trained and evaluated on four distinct datasets, incorporating both real-world empirical measurements and publicly available repositories. This diverse data exposure, coupled with data augmentation techniques, ensured that the model learned generalizable features rather than overfitting specific datasets. Additionally, k-fold cross-validation was employed to assess the model's stability

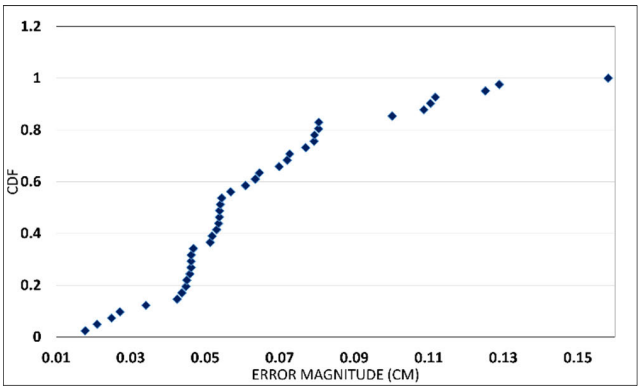


FIGURE 5. The CDF of the overall errors.

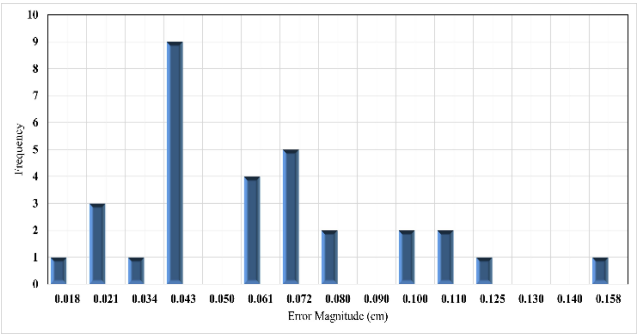


FIGURE 6. Distribution of aggregate errors.

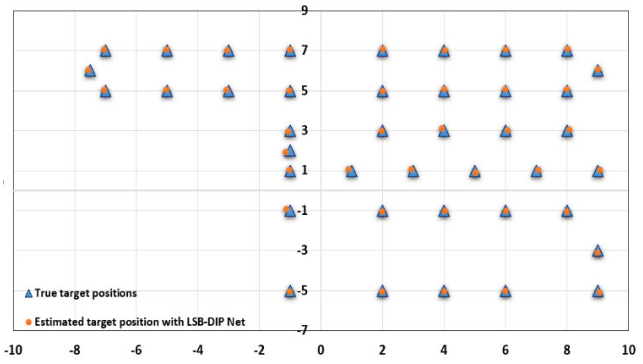


FIGURE 7. True vs estimated positions based on LSB-DIP Nets.

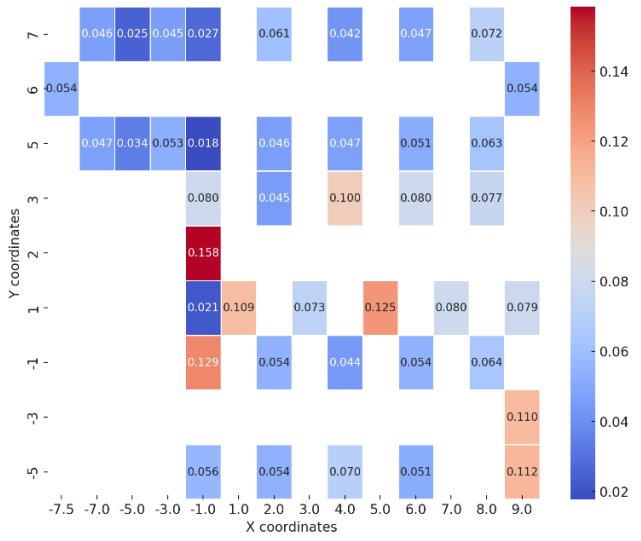


FIGURE 8. Error distributions heatmap.

and performance across various data partitions, providing a reliable estimate of its generalization capability. Furthermore, the LSB-DIP Net's architecture integrated MaxPooling1D layers to reduce feature dimensionality and enhance translational invariance, as well as L2 regularization to constrain weight magnitudes and prevent excessive parameter growth. The model's consistent performance across independent test datasets, as presented in Table 4, confirms its strong generalization ability and the successful mitigation of overfitting. By integrating dataset diversity, data augmentation, cross-validation, and architectural regularization, the LSB-DIP Net demonstrates robustness and adaptability across diverse real-world scenarios.

The LSB-DIP Net system was further validated using real-world applications, including a simulated 90-meter trajectory in a 20 m $\times$ 20 m indoor environment containing thick walls and office furniture. The CDF was employed to evaluate the reliability of the positioning system by quantifying the proportion of errors falling within specified tolerance thresholds. As illustrated in Fig. 5, the CDF depicts the cumulative probability of overall errors, thereby providing a comprehensive representation of the error distribution. This function facilitates a clear understanding of the error distribution characteristics. This metric reflects the system's performance by indicating that the majority of errors remain within a manageable range. The curve's profile reveals a concentration of smaller errors, exhibiting a steep gradient between 0.03 and 0.06 units. This suggests a substantial proportion of errors reside within this interval. The CDF further confirms the prevalence of errors within the lower range, with only a minor percentage exceeding 0.1 units. This characteristic indicates a high degree of consistency and reliability in the positioning system's performance. While a limited number of larger errors were observed, potentially attributable to the inherent complexities of indoor environments, the system's performance nonetheless surpasses the state-of-the-art benchmarks reported in the existing literature.

Histograms are also shown to visualize the distribution of error magnitudes in the (x, y) plane (Fig. 6). The histogram of overall error magnitudes reveals a concentration of values between 0.04 and 0.06 units, with a peak frequency observed at approximately 0.05 units. This narrow distribution suggests consistent system performance. The histogram further demonstrates a relatively confined spread of errors between 0.034 and 0.072 units, with a lower frequency of occurrences outside this interval. This characteristic indicates that the majority of errors are of small magnitude, reflecting reliable performance. Larger error magnitudes, such as those observed around 0.125 and 0.158 units, are infrequent, and potentially indicative of

TABLE 5. Comparison of recent results for different IPSs using UWB technology.

Study	Positioning Accuracy (cm)	NLOS & Environment Handling	Generalizability to Real-World Environments	Computational Complexity	Strengths	Limitations
LSB-DIP Net	<17, with an MSE of 0.0051m², RMSE of 0.0716m	Yes, precisely identification with multiple NLOS conditions (LOS, MP, soft-NLOS, hard-NLOS)	High	Low	High accuracy, robust NLOS identification, low complexity, strong generalization	-Requires investigation in a more large-area environment -Utilize an alternative obstacle dataset to establish a more comprehensive framework.
[25]	±50	Yes, Limited to specific scenarios (pocket, bag, and hand)	Moderate (needs additional data)	Moderate	Good NLOS mitigation, real-time error correction	Limited to specific NLOS scenarios, struggles in complex environments
[26]	±50	YES, Limited to binary channel (LOS, NLOS)	Moderate (focused on controlled simulation-)	Moderate	Handles non-linear state estimation effectively	Dependent on noise characteristics, limited real-world testing
[27]	~25	No	Moderate (Generalization may compromise accuracy in challenging environments.)	Moderate	Simplified system with good accuracy, avoiding NLOS identification	Lacks NLOS handling, and highly sensitive random anchor placement. Accuracy could be compromised by overgeneralization.
[29]	32	No	Moderate (limited on specific Simulation Environments)	Moderate	Autonomous anchor selection with RL	Idealized simulation conditions, lacks robustness in real-world environments
[31]	MAE (14.9, 20.2) STD (7.1, 14.8)	Partial, Limited to binary channel (LOS, NLOS))	Moderate (industrial setting)	Moderate	High accuracy in industrial settings	High computational requirements depend on specific metrics. Evaluated in a controlled industrial, possesses challenges for generalizing effectively to diverse or more complex indoor environments.
[28]	~50	Yes	Low (controlled settings)	High	Dynamic anchor selection for improved accuracy	High computational overhead, limited dataset applicability
[32]	> 100 (in some NLOS areas)	Partially (based on channel quality metrics)	Moderate (requires recalibration)	Moderate	Significant RMSE reduction using in specific cases	Requires recalibration, computationally intensive -Noise Sensitivity
[30]	6-108	No	Limited to specific settings	Moderate	Real-Time Anchor Node Selection	-Limited Evaluation Scenarios -Generalization Concerns
[33]	78.2	No	Moderate (it relies on mode values from prior distributions)	Moderate	Utilizing smartphone inertial sensors with LSTM-based trajectory updates in dynamic environments achieved accuracy in physical activity (97.9%) and action unit (95.5%) recognition	-Limited Evaluation Scenarios -Generalization Concerns
[34]	-LSTM-based WLS (74).	No	Moderate (focused on controlled simulation-)	Moderate	Using LSTM-based CIR error prediction with eight anchors,	-Dependence on the distribution and

TABLE 5. *(Continued.)* **Comparison of recent results for different IPSs using UWB technology.**

	-LSTM-based WRLS (86). -LSTM-based LLS (>100)				positioning errors of 0.74m (WLS), 0.86m (WRLS), and >1m (LLS) were achieved.	quantity of anchors used. -Limited Dataset scope
[36]	40	Yes	Moderate (Generalization may compromise accuracy in challenging environments.)	Moderate, with consumption time of 11.05 ms per step	Adaptive filtering for improved robustness, deep learning for motion estimation, real-time processing capability	-Computationally intensive due to Firefly optimization. -Requires a well-trained deep learning model for step length estimation; -Performance degradation in highly cluttered environments
[37]	29 based LOS, and 37 based NLOS	Yes	Moderate	high	real-time localization capability, NLOS compensation	-Higher computational demands. -Best performance relies on multi-anchor setup. -Residual inaccuracies in highly reflective environments.
[38]	MSE of 0.46 m^2	No	Moderate (Generalization may compromise accuracy in challenging environments.)	Moderate	Simplified system with good accuracy, avoiding NLOS identification	-Limited scalability for large environments. -Dependent on proper anchor distribution. -Performance in highly dynamic settings remains untested.
[39]	15 (based on 4 anchors of LOS), 10 (based on 4 anchors of LOS, and 2 anchors of NLOS)	No, select the anchor in LOS	High accuracy was achieved under restrictive LOS conditions with 4 anchors, limiting the general applicability of the results	High	Autonomous anchor selection with PSO	-Higher computational cost due to iterative PSO optimization. -Performance depends on proper LOS anchor placement. -Scalability in large and highly dynamic environments remains untested.
[40]	RMSE of 40 to 51	No, Does not explicitly address NLOS	Tested in a controlled indoor setup and requires further validation in large-scale and complex environments to ensure the generality.	Moderate	Improves localization accuracy compared to standard TDoA-based methods; Real-time position tracking capabilities.	-Performance dependent on anchor placement. -No explicit NLOS mitigation strategy.
[41]	≤ 50	Yes	Moderate (Generalization may compromise accuracy in challenging environments.)	Moderate	Good accurate NLOS classification (97.6% hallway, 96.6% office); Significant improvement in positioning accuracy over traditional Kalman Filtering;	-Localization accuracy degrades in 3D multi-room environments. -Not optimized for continuous real-time tracking.
[35]	In LOS environment (7)	No	Low (controlled LOS settings)	Moderate	This hybrid approach combines UWB's precision with LSTM's temporal modeling to mitigate LOS issues, achieving ~7 cm mean error.	-Limited generalization, focus on LOS challenges environment. -Dependence on the quantity of anchors used. -High LSTM training time may hinder real-time deployment.

specific challenging scenarios encountered during testing. The observed concentrated peak and narrow spread of the error distribution underscore a high level of accuracy and consistency in the system's performance. The prevalence

TABLE 6. List of abbreviations.

Abbreviation	Definition
UWB	Ultra-wideband
TWR	Two-way-ranging
LSB-DIP Net	Lightweight sequential branching deep network for dynamic indoor positioning
NLOS	Non-line-of-sight
MSE	Mean squared error
RMSE	Root mean squared error
STD	Standard deviation
IPSs	Indoor positioning systems
RSSI	Received signal strength indication
TOF	Time-of-flight
TOA	Time of arrival
TDOA	Time difference of arrival
AOA	Angle of arrival
AS	Anchor selection
LOS	Line-of-sight
soft-NLOS	Soft non-line-of-sight
hard-NLOS	Hard non-line-of-sight
MP	Multipath
DL	Deep learning
CIR	Channel impulse response
WAKF	Weighted adaptive Kalman filter
EKF	Extended Kalman Filter
LLSE	Linearized least squares estimation
WCE	Weighted centroid estimation
MLE	Maximum likelihood estimation
LLS	Linearized least squares
WGDOP	Weighted Geometric Dilution of Precision
MS	Mobile station
LS	Least Squares
WLS	Weighted Least Squares
JSRL	jump-start reinforcement learning
JUNO	Jump-start RL-based Uwb NOde selection
BAST	Best Anchor Selection for Trilateration
ED	Euclidean distance
DOP	Dilution of precision
OS-CFAR	Ordered statistical constant false alarm rate
RSL	Received signal strength
FSL	First path signal level
PSC	Preamble symbol count
ML	Machine learning
RNNs	Recurrent neural networks
Max Error	Maximum Absolute Error
STD	Standard Deviation of Error
CDF	Cumulative Distribution Function of Error
PRF	Pulse repetition frequency
LSTM	Long Short-Term Memory
AUs	Action units
AREKF	Adaptive robust extended Kalman filter
SVM-RFE	SVM- recursive feature elimination
PCC	Pearson correlation coefficient
PSO	Particle swarm optimization
ELLS	Enhanced LLS
DPGMM	Dirichlet process gaussian mixture model
MAP	Maximum A posteriori
INS	Inertial navigation systems
AEKF	Adaptive extended Kalman filter

of errors within a manageable range contributes to reliable positioning.

The maximum observed error magnitudes were 0.1311 units in the x-axis direction and 0.1194 units in the y-axis direction. These results underscore the system's accuracy and robustness, particularly in challenging environments. The results demonstrate a minimal position error of less than 0.17 m in a challenging hard-NLOS environment. Notably, the LSB-DIP Net system demonstrates superior performance compared to state-of-the-art techniques, establishing it as a valuable solution for complex and demanding positioning scenarios.

The scatter plot of Fig. 7 provides a visual representation of the performance of the LSBDI NETs algorithm in estimating target positions within an indoor environment. The blue markers denote the true target positions, while the

red markers correspond to the estimated positions generated by the algorithm. In general, the estimated positions align closely with the true positions, highlighting the accuracy of the algorithm. However, noticeable discrepancies are observed in certain instances, which can be attributed to the significant impact of multiple selected anchors with hard NLOS or sparse outlier data points, which degrade the algorithm's performance.

Fig. 8 illustrates spatial variations in estimation errors, where brighter regions correspond to higher error magnitudes and darker areas indicate regions of greater positional accuracy. High-error regions are predominantly associated with challenging environmental conditions, potentially resulting from the selection of multiple anchors subjected to severe NLOS interference or the presence of sparse, outlier data points, both of which adversely impact the algorithm's performance. In contrast, regions with low errors are indicative of favorable conditions, characterized by the selection of multiple optimally placed anchors, contributing to enhanced accuracy in the position estimation process.

The localization error distribution reveals that the highest errors, exceeding 0.12 meters, are concentrated in specific regions within the test environment. The maximum recorded error reaches 0.158 meters (approximately in $(-3,2)$), while other high-error zones exhibit errors exceeding 0.10 meters. These regions are predominantly characterized by severe NLOS conditions, wherein all anchor signals are classified as hard-NLOS. This results in a ranging bias due to excessive signal delay. Additionally, these areas are susceptible to multipath effects, where signal reflections from nearby surfaces introduce interference and distance fluctuations, further degrading localization accuracy. Moderate localization errors, ranging between 0.08 and 0.10 meters, are observed in several locations across the test environment. In contrast, the majority of the experimental area exhibits low localization errors, ranging between 0.02 and 0.06 meters. These regions comprise a mix of LOS and NLOS conditions, including areas with severe NLOS cases (hard-NLOS). The relatively low error levels in these areas demonstrate the robustness of the LSB-DIP Net in effectively distinguishing between different signal propagation conditions, even in challenging indoor environments. This underscores the model's ability to mitigate NLOS-related errors, thereby enhancing localization accuracy and reliability in complex real-world scenarios.

The model achieves exceptional performance across diverse datasets and experimental scenarios. Its low computational complexity and adaptability to real-world challenges establish it as a benchmark solution for anchor-channel selection and NLOS identification in UWB positioning systems. Additionally, the proposed algorithm was benchmarked against the state-of-the-art methods. As shown in Table 5 in Appendix, the LSB-DIP Net-based approach outperforms benchmarks in both positioning accuracy and computational efficiency, highlighting its practicality for dynamic and resource-constrained environments. These findings underscore its transformative potential for applications such as emergency response and resource-constrained indoor localization.

VII. CONCLUSION

This paper presents a lightweight sequential branching deep network for dynamic indoor positioning (LSB-DIP Net), designed to enhance UWB indoor positioning by integrating multi-scale feature extraction, sequential learning, and advanced activation functions with the conventional linearized least squares method. The proposed model demonstrates robust, accurate, and dynamic positioning capabilities in real-time scenarios, effectively addressing challenges such as multi-class NLOS conditions and the dynamic selection of optimal anchor channel quality.

The LSB-DIP Net achieved superior performance metrics, including an identification of a multi-optimal anchor channel with an accuracy of 99.44% and perfect precision, recall, and F1-scores. Notably, the model demonstrated impressive efficiency with a training time of only 1 second per epoch. In a realistic dynamic setup, the algorithm exhibited robust performance with an MSE of 0.0051m^2, RMSE of 0.0716m, and an STD of 0.0303m. The maximum positional errors observed were 0.1311 units on the x-axis and 0.1194 units on the y-axis, corresponding to a minimal positioning error of less than 0.17 m, even in challenging hard-NLOS environments. These results confirm the reliability and potential of LSB-DIP Net for real-world applications in communication and navigation systems requiring prompt and accurate signal classification.

Future work will focus on extending the scalability of the LSB-DIP Net to larger and more complex datasets and assessing its adaptability in diverse and variable signal environments. Further investigations will include optimizing a new approach that gathers static and dynamic algorithms in one design, benchmarking the model against emerging machine learning methods, and enhancing its interpretability to improve robustness against adversarial attacks. Also, future work should prioritize the exploration of hardware acceleration methodologies to enable real-time, low-power deployment within resource-constrained environments. The implementation of lightweight deep learning architectures optimized for both accuracy and computational efficiency is crucial for the practical integration of UWB-based localization into mobile and embedded systems.

APPENDIX

See Tables 5 and 6.

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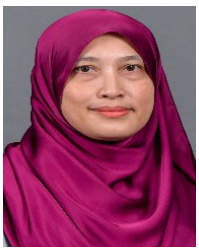
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